Robust Adaptive Kolmogorov-Arnold Neural Control

Adilkhan Salkimbayev and Khawaja S. Haider

School of Information Technology and Engineering, Kazakh-British Technical University, Almaty, Kazakhstan

Abstract

Reliable real-time control of non-minimum phase systems remains a bottleneck for embedded platforms due to the computational burden of online optimization, cumbersome architecture and insufficient stability guarantees in the presence of model mismatch. We propose a Lipschitz-continuous adaptive (LCA) architecture using recursive least squares (RLS) to update Kolmogorov-Arnold networks (KANs) for rapid, robust, and interpretable control of control-affine multivariable systems with a left-invertible input map. The robustness claim is established via Lyapunov-based analysis, proving input-to-state stability with respect to the lumped approximation-and-disturbance error and global uniform ultimate boundedness of the closed-loop error state. A high-fidelity quadruple-tank benchmark under adversarial parameter drift is used as a representative instance, evaluated over a 50-seed Monte-Carlo against linear time-varying model predictive control (LTV-MPC) and a radial-basis-function (RBF) adaptive baseline. The proposed controller matches the aggregate tracking of both while engaging the cross-coupled actuation that LTV-MPC under-serves, holding the coupled, non-minimum-phase tank approximately 23% closer to setpoint than LTV-MPC. Additionally, the controller produces the smoothest command of all four controllers and lower actuator total variation than both the RBF baseline and a prediction-horizon LTV-MPC, combined with the lowest median per-step cost at a branch-free, worst-case-bounded evaluation. Because the controller is distilled into an explicit symbolic (polynomial) regressor, the resulting control law is directly inspectable, in contrast to optimization-based or uninterpretable neural controllers. The architecture is therefore particularly suited to resource-constrained real-time embedded platforms (e.g., Cortex-M-class microcontrollers), where bounded, deterministic per-step evaluation is critical.

Key words: Adaptive control, model predictive control, Kolmogorov-Arnold networks, neuro-symbolic control, real-time control, interpretable neural networks, non-minimum phase systems

1 Introduction

Model predictive control (MPC) and robust control strategies have demonstrated exceptional performance in numerical benchmarks, displaying extraordinary resilience and disturbance rejection (Jerez et al., 2014). These results have extensively validated the theoretical stability guarantees established in foundational works (Mayne et al., 2000; Borrelli et al., 2017). However, the computational requirements for such optimization-based techniques have impeded their widespread deployment in industrial plants. Despite evolution in quadratic program (QP) solvers (Stellato et al., 2020) and the democratization of convex optimization tools (Boyd and Vandenberghe, 2004; Diamond and Boyd, 2016), a stalemate has emerged: industrial sectors remain reliant on PID control, largely unable to adopt MPC due to the hardware requirements of online optimization. Thus, a need for a computationally efficient nonlinear controller persists.

Email addresses: a_salkimbayev@kbtu.kz (Adilkhan Salkimbayev), s.khawaja@kbtu.kz (Khawaja S. Haider).

Recent data-driven approaches have sought to replace resource-intensive online optimization with learnable architectures. While methods like SINDYc (Brunton et al., 2016) improve interpretability through sparse identification, the broader adoption of neural control has been stifled by the intractable structure and uninterpretable nature of standard multi-layer perceptrons (MLPs) (Hose et al., 2024). By contrast, Kolmogorov-Arnold networks (KANs) provide an alternative: by utilizing the Kolmogorov-Arnold theorem of universal function approximation (Kolmogorov, 1957; Arnold, 1957), KANs are capable of providing a tractable symbolic structure, possessing faster scaling laws than MLPs (Liu et al., 2024). As a result, KANs enabled research on interpretable neural network-based system identification and control (Cruz et al., 2025; Kim et al., 2025).

In previous work (Salkimbayev, 2026), the feasibility of deploying static KANs for feedback control was established, demonstrating preservation of high fidelity of the control law with interpretable structure, which enabled a forensic audit of KAN's output and relative simplicity of the control law. However, the process relied on offline training, leaving the controller vulnerable to unmodeled dynamics and parameter drift once deployed and providing no formal guarantees of robustness.

Online function-approximation adaptive control is a mature paradigm: radial-basis-function (RBF) networks (Sanner and Slotine, 1991) and feedforward neural networks (Lewis et al., 2020; Ge et al., 2013) have long been used as in-loop regressors whose weights adapt online, with Lyapunov-based boundedness guarantees rooted in classical robust adaptive control (Ioannou and Sun, 1996; Astrom, 1987; Narendra and Annaswamy, 2012; Krstic et al., 1995). The distinction between the proposed controller and this paradigm is not adaptation per se but the form of the regressor: where RBF/MLP schemes adapt a black-box basis, the proposed controller adapts the linear weights of an interpretable, symbolically distilled KAN basis using recursive least squares (RLS) (Plackett, 1950), retaining the auditability of the static KAN control law while recovering robustness to drift. This positions the contribution between black-box neural-adaptive control and sparse-identification approaches such as SINDYc (Brunton et al., 2016).

In this article, the Lipschitz-continuous adaptive RLS-KAN (LCA-RLS-KAN) is proposed. The contributions of this paper are threefold: first, RLS is introduced as an update law for the KAN-distilled polynomials that operates online; second, a Lyapunov-based theorem establishing input-to-state stability (ISS) and global uniform ultimate boundedness (GUUB) of the tracking and weight errors, with the (separately, a priori bounded) integral state handled via a leaky integrator confined to an explicit band, is provided, with corollaries quantifying practical tracking accuracy in the ideal case and robustness under lack of persistent excitation (PE); third, a 50-seed Monte-Carlo study on a high-fidelity quadruple-tank benchmark shows that the controller matches LTV-MPC and a black-box RBF adaptive baseline on aggregate tracking while engaging the cross-coupled, NMP actuation that MPC under-serves, with the lowest actuator total variation of all four controllers and the lowest median per-step cost at a worst-case-bounded, branch-free evaluation — architectural advantages for embedded deployment.

2 Related Work

Function-approximation adaptive control. The closest methodological relatives adapt an in-loop function approximator online under a Lyapunov boundedness guarantee: RBF networks (Sanner and Slotine, 1991) and feedforward/recurrent neural networks for robotic and nonlinear systems (Lewis et al., 2020; Ge et al., 2013). These supply an uninterpretable regressor whose internal units carry no physical meaning. The proposed method retains the adaptive-update machinery of this lineage but replaces the regressor with an interpretable, symbolically distilled KAN basis; the controlled RBF baseline of Sec. 5.2 is precisely a representative of this class, so that comparison isolates the effect of the basis itself.

Table 1
Positioning relative to representative control approaches. ✓: yes; ∼: partial; ×: no.

Approach	**Online adapt.**	**Interp. law**	**Stability cert.**	**Determ. per step**
LTV / nonlinear MPC	∼	×	✓	×
Approximate MPC (MLP) (Hose et al., 2024)	×	×	∼	✓
RBF/MLP adaptive (Sanner and Slotine, 1991; Ge et al., 2013)	✓	×	✓	✓
SINDYc (Brunton et al., 2016)	×	✓	∼	✓
Static symbolic KAN (Salkimbayev, 2026)	×	✓	×	✓
LCA-RLS-KAN (ours)	✓	✓	✓	✓

Robust and adaptive control foundations. The robustification employed here — exponential forgetting with bounded-gain covariance handling (Fortescue et al., 1981), an optional σ-modification (Sec. 4.7), and leakage or anti-windup on the integral state — is classical (Ioannou and Sun, 1996; Astrom, 1987; Narendra and Annaswamy, 2012; Krstic et al., 1995; Slotine and Li, 1991). Input-to-state stability (Sontag, 1989; Khalil and Grizzle, 2002) is the property certified in Sec. 4.6.

Learning-based and approximate MPC. A parallel line replaces online optimization with a learned surrogate of an MPC policy, typically an MLP (Hose et al., 2024), or identifies sparse governing dynamics for control (Brunton et al., 2016). Such surrogates inherit the opacity of their underlying networks. Distillation into a symbolic law (Sec. 4.4) instead yields an inspectable controller; the present contribution is to make that distilled law adaptive and to equip it with a stability certificate.

Kolmogorov–Arnold networks and neuro-symbolic control. KANs (Liu et al., 2024), grounded in the Kolmogorov–Arnold representation theorem (Kolmogorov, 1957; Arnold, 1957), have recently been applied to interpretable system identification (Cruz et al., 2025) and constrained control (Kim et al., 2025). This work exploits their distillability as the route to a symbolic, online-adaptive control law.

The quadruple-tank benchmark. Johansson's process (Johansson, 2000) is a standard multivariable, adjustable-zero, NMP testbed (Skogestad and Postlethwaite, 2005), studied under model-based (Gatzke et al., 2000) and distributed/decentralized MPC (Alvarado et al., 2011); it is used here as a representative instance of the control-affine class of Sec. 3.1. Table 1 positions these approaches on the axes that matter for embedded deployment.

Table 2
Notation used in the stability analysis.

Symbol	**Meaning**
x	plant state (here, tank levels $h_{1\ldots4}$)
$e = x - x_{\mathrm{ref}}$	tracking error
$\hat{\theta},\ \theta^*$	estimated / ideal linear KAN weights
$\tilde{\theta} = \hat{\theta} - \theta^*$	weight estimation error
$\hat{d},\ d^*$	integral state / unknown constant bias
$\tilde{d} = \hat{d} - d^*$	integral-state error (bounded a priori by construction)
$\zeta = [\, e^T\ \tilde{\theta}^T\,]^T$	composite state vector
$\Phi_{\mathrm{KAN}}(x)$	distilled nonlinear regressor vector
$\delta(x)$	lumped bounded disturbance (see text)

3 Problem Formulation

3.1 System Class and Notation

The controller is developed for the class of control-affine multivariable systems

$$\dot{x} = f(x) + B\,u, \qquad x \in \mathbb{R}^n,\ \ u \in \mathbb{R}^m, \tag{1}$$

where $f : \Omega \to \mathbb{R}^n$ is continuous on a compact operating set $\Omega \subset \mathbb{R}^n$ and $B \in \mathbb{R}^{n\times m}$ has full column rank (left-invertible), so that the left pseudoinverse $B^+ = (B^TB)^{-1}B^T$ satisfies $B^+B = I$. The plant may be NMP and may be underactuated ($m < n$); the residual $r_{\mathrm{ua}} := (BB^+ - I)(\cdot)$ arising from underactuation is treated as a bounded disturbance (Lemma 1). The control objective is tracking of a feasible, bounded reference $x_{\mathrm{ref}}(t)$ with bounded $\dot{x}_{\mathrm{ref}}(t)$, with all internal signals remaining bounded under non-vanishing model mismatch. The quadruple-tank process of Sec. 3.2 is a representative instance of (1).

Throughout, $\|\cdot\|$ is the Euclidean norm and $\|\cdot\|_\infty$ the essential-supremum norm; for $M = M^T \succ 0$, $\lambda_{\min}(M), \lambda_{\max}(M)$ denote its extreme eigenvalues and $\|v\|_M^2 := v^TMv$. The error variables used in the analysis are summarized in Table 2. The Lyapunov analysis (Sec. 4.6) is conducted over the composite state $\zeta = [\, e^T\ \tilde{\theta}^T\,]^T$; the integral-state error $\tilde{d}$ is tracked separately and bounded directly by construction (Lemma 2) rather than through ζ.

3.2 Benchmark Instance: Quadruple-Tank Process

To compare the controllers, Johansson's quadruple-tank process is used — a nonlinear multivariable benchmark consisting of four interconnected liquid tanks and two control inputs, exhibiting strong cross-coupling and NMP behavior (Johansson, 2000). It is an instance

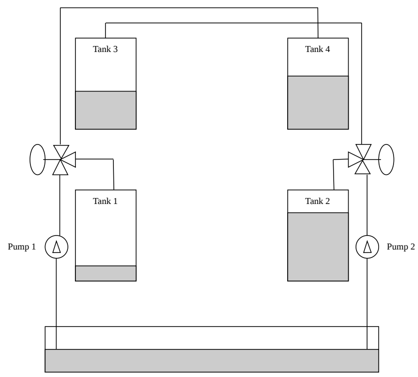

Fig. 1. **Quadruple-tank process.** Four interconnected tanks driven by two pumps; each pump i splits its flow by valve ratio γ_i between a lower tank and the diagonally opposite upper tank, producing the strong cross-coupling and, for $0 < \gamma_1 + \gamma_2 < 1$, the non-minimum-phase zero.

of (1) with $n = 4$, $m = 2$, and has since become a standard benchmark for multivariable, decentralized, and distributed control design (Gatzke et al., 2000; Alvarado et al., 2011). The process can be configured to minimum-phase or NMP regimes, enabling a controlled assessment of performance degradation and robustness challenges in adversarial control settings (Johansson, 2000; Skogestad and Postlethwaite, 2005).

The quadruple-tank model follows these equations (Johansson, 2000):

$$\begin{aligned}
\frac{dh_1}{dt} &= -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1}\sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1}u_1 \\
\frac{dh_2}{dt} &= -\frac{a_2}{A_2}\sqrt{2gh_2} + \frac{a_4}{A_2}\sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2}u_2 \\
\frac{dh_3}{dt} &= -\frac{a_3}{A_3}\sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3}u_2 \\
\frac{dh_4}{dt} &= -\frac{a_4}{A_4}\sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4}u_1
\end{aligned} \tag{2}$$

In (2), tank and outlet cross-sectional areas $A = [28, 32, 28, 32]$, $a = [0.071, 0.057, 0.071, 0.057]$ are in cm^2; levels h_i (cm), pump voltages u_i (V), pump gains $k_1 = 3.14, k_2 = 3.29$ (cm^3/Vs), valve ratios are $\gamma_1 = 0.43, \gamma_2 = 0.34$, and $g = 981$ cm/s^2. The plant schematic is shown in Fig. 1.

3.3 Control Objective

The plant configuration follows Johansson's NMP setup, described in (Johansson, 2000). The target state is set as $[10.0, 10.0, 2.0, 2.0]$ for tanks 1, 2, 3 and 4 respectively. In addition to the challenge of NMP control, all controllers are subjected to structured adversarial perturbations: slow parametric drift of valve coefficients (modeling wear and aging), step-wise parameter deviations, and bounded external disturbances. These perturbations are non-vanishing, bounded and positioned outside of the nominal modeling assumptions typically required for LTV-MPC stability guarantees, while remaining physically plausible in industrial fluid systems. All perturbations are applied identically to all control systems, with no controller-specific tuning or compensation.

Performance is assessed over a 50-seed Monte-Carlo campaign (Sec. 5): each seed draws independent process- and measurement-noise realizations under an identical adversarial schedule,

- coefficient γ_1 of the first pump degrades at a constant rate of $-0.0011/\mathrm{s}$;
- coefficient γ_2 of the second pump undergoes a step-wise 20% drop at $t = 25\,\mathrm{s}$.

A separate single-realization stress test (Sec. 5) additionally injects a $2.0\,\mathrm{cm}$ step disturbance into tank 2 at $t = 30\,\mathrm{s}$, on top of the drift schedule, to expose cross-coupled recovery under model mismatch.

4 Controller Design & Stability Analysis

4.1 Preliminaries: Kolmogorov–Arnold Representation

The architecture rests on the Kolmogorov–Arnold representation theorem (Kolmogorov, 1957; Arnold, 1957): any continuous $g : [0,1]^n \to \mathbb{R}$ admits the finite superposition

$$g(x) = \sum_{q=0}^{2n} \Phi_q\Big(\sum_{p=1}^{n} \phi_{q,p}(x_p)\Big), \tag{3}$$

with continuous univariate inner and outer functions $\phi_{q,p}, \Phi_q$. A KAN (Liu et al., 2024) parameterizes these univariate maps as learnable splines placed on the network edges, while nodes perform summation; this is the structural dual of an MLP, in which fixed nonlinearities sit on nodes and learnable weights on edges. After offline training on input–output data of the plant nonlinearity, the trained univariate maps are distilled into a fixed, finite dictionary of basis functions, collected in the regressor $\Phi_{\mathrm{KAN}}(\cdot)$, so that the control-relevant nonlinearity of (1) is represented as a linear-in-parameters model,

$$f(x) = \Phi_{\mathrm{KAN}}(x)^T\theta^* + \varepsilon(x), \tag{4}$$

where θ^* are the (unknown, constant) ideal weights and ε is the residual approximation error. Only the linear weights θ are adapted online; the distilled basis is frozen. For the quadruple-tank instance, the inputs to Φ_{KAN} are the measured/estimated tank levels and the regressor reproduces the square-root outflow and cross-coupling terms of (2); the distilled law for u_1 reduces to the explicit algebraic form reported in Sec. 4.5.1.

Proposition 1 (Regularity of the distilled basis) *The distilled regressor $\Phi_{\mathrm{KAN}} : \Omega \to \mathbb{R}^p$ is C^∞ and, on the compact operating set Ω, both bounded and Lipschitz: the constants $\Phi_{\max} := \sup_{x\in\Omega}\|\Phi_{\mathrm{KAN}}(x)\|$ and $L_\Phi := \sup_{x\in\Omega}\|\partial\Phi_{\mathrm{KAN}}/\partial x\|$ are finite, and $\|\Phi_{\mathrm{KAN}}(x) - \Phi_{\mathrm{KAN}}(y)\| \le L_\Phi\|x-y\|\ \forall\, x, y \in \Omega$. For the quadruple-tank instance $p = 10$: seven affine state/error features, two quadratic NMP-coupling features, and a bias term, as instantiated in* (13)–(14)*.*

Proof. By the symbolic distillation (13)–(14), every component of Φ_{KAN} is either an affine function of the normalized features, the square of an affine function (the quadratic NMP-coupling terms), or a constant; hence each is a multivariate polynomial of degree at most two and $\Phi_{\mathrm{KAN}} \in C^\infty(\mathbb{R}^n, \mathbb{R}^p)$. Input normalization confines the operating envelope to the compact box Ω, on which the continuous map Φ_{KAN} attains its supremum (Weierstrass), so $\Phi_{\max} < \infty$. The Jacobian $\partial\Phi_{\mathrm{KAN}}/\partial x$ is affine, hence continuous and bounded on Ω; since the box Ω is convex, the mean-value inequality gives the stated Lipschitz bound with $L_\Phi < \infty$. Boundedness therefore needs no separate enforcement in implementation: it is a property of evaluating a fixed polynomial dictionary on a compact domain. □

Proposition 2 (Boundedness of ideal weights) *Define ideal weights as $L^2(\Omega)$ projection of the plant nonlinearity onto the frozen basis,*

$$\theta^* := \arg\min_{\theta\in\mathbb{R}^p} \left\|f - \Phi_{\mathrm{KAN}}^T\theta\right\|_{L^2(\Omega)}^2, \tag{5}$$

which makes precise the ideal weights of (4) *and pins $\varepsilon = f - \Phi_{\mathrm{KAN}}^T\theta^*$ as the $L^2(\Omega)$-orthogonal residual — the model error that no choice of weights can cancel. Let the distilled features be linearly independent on Ω, equivalently $\lambda_{\min}(G) > 0$ for the Gram matrix $G := \int_\Omega \Phi_{\mathrm{KAN}}\Phi_{\mathrm{KAN}}^T\,dx$. Then θ^* is unique and bounded:*

$$\begin{aligned} \theta^* &= G^{-1}\int_\Omega \Phi_{\mathrm{KAN}}\, f\, dx, \\ \|\theta^*\| &\le \frac{\Phi_{\max}\sup_\Omega |f|\,|\Omega|}{\lambda_{\min}(G)} =: \theta_{\max}. \end{aligned} \tag{6}$$

For the vector plant the bound applies to each output channel, all sharing the Gram G.

Proof. The quadratic objective (5) has normal equations $G\theta^* = \int_\Omega \Phi_{\mathrm{KAN}} f\,dx$. Under $\lambda_{\min}(G) > 0$ the symmetric Gram G is positive definite, so the minimizer $\theta^* = G^{-1}\int_\Omega \Phi_{\mathrm{KAN}} f\,dx$ is unique and ε is $L^2(\Omega)$-

orthogonal to the basis. Hence

$$\begin{aligned}\|\theta^*\| &\le \|G^{-1}\|_2 \Big\| \int_\Omega \Phi_{\mathrm{KAN}} f \, dx \Big\| \\ &\le \frac{1}{\lambda_{\min}(G)} \int_\Omega \|\Phi_{\mathrm{KAN}}\| \, |f| \, dx \le \frac{\Phi_{\max} \sup_\Omega |f| \, |\Omega|}{\lambda_{\min}(G)},\end{aligned} \tag{7}$$

using $\|G^{-1}\|_2 = \lambda_{\min}(G)^{-1}$, boundedness of f (f continuous on the compact Ω, plant class (1)), and $\Phi_{\max} < \infty$ (Proposition 1). The hypothesis is a pure conditioning requirement: were $\|\theta^*\|$ unbounded the normal equations would force $\lambda_{\min}(G) = 0$, i.e. a nonzero v with $\int_\Omega |\Phi_{\mathrm{KAN}}^T v|^2 dx = 0$ and hence a linear dependence $\Phi_{\mathrm{KAN}}^T v \equiv 0$ among the features — contradicting their independence. $\square$

Remark. In practice the constant is read directly off the frozen distilled coefficients: $\|\theta^*\| \approx 12.3$ for the primary channel (2.9 for the secondary), so $\theta_{\max} \approx 12.3$ — mirroring how the integral bound $\bar{d}$ is read off logged data (Lemma 2). The conditioning $\lambda_{\min}(G) > 0$ holds on the reference-varying distillation set of Algorithm 1, where the empirical Gram $G_N = \frac{1}{N}\sum_i \Phi_{\mathrm{KAN}}(\xi_i)\Phi_{\mathrm{KAN}}(\xi_i)^T$ is full rank. At a single regulation setpoint, by contrast, the error features collapse onto the level features ($x_5 = c_1 - x_1$, $x_7 = c_3 - x_3$, $x_8 = c_4 - x_4$) and G_N drops to rank seven, so PE (persistent excitation) is lost — precisely the regime in which the covariance bounding of the RLS gain (Assumption 3) and Corollary 2 operate. The same feature-conditioning $\lambda_{\min}$ thus wears two hats: bounded below on the distillation set it makes θ^* well-posed, while its degeneracy at the operating point is what renders the covariance bound of Assumption 3 load-bearing rather than decorative.

4.2 Standing Conditions

The following conditions are used; each is referenced at the point of use.

Assumption 1 (Input map) *B has full column rank at the operating point, so $B^+B = I$.*

Lemma 1 (Lumped disturbance bound) *The lumped disturbance $\delta(t) \triangleq \varepsilon_{\mathrm{tv}}(x) + r_{\mathrm{ua}}(t) + B\, r_{\mathrm{slew}}(t) + r_{\mathrm{ukf}}(t)$, collecting the time-varying approximation error, the underactuation residual, the actuation-realization residual of Sec. 4.5.1, and the state-estimation residual, is uniformly bounded, $\delta \in \mathcal{L}_\infty$. The constant component of the approximation error ε is denoted d^*, with $\|d^*\| \le d_{\max}$.*

Proof. Each of the four components is bounded on the compact set Ω. (i) The approximation error $\varepsilon = f - \Phi_{\mathrm{KAN}}^T\theta^*$ is continuous on Ω — f is continuous on the compact Ω (plant class (1)) and Φ_{KAN} is bounded (Proposition 1) and θ^* bounded (Proposition 2) — hence bounded, so its time-varying part $\varepsilon_{\mathrm{tv}} = \varepsilon - d^*$ is bounded. (ii) The underactuation residual $r_{\mathrm{ua}} = (BB^+ - I)(\cdot)$ is a fixed projector applied to bounded arguments on Ω, hence bounded. (iii) The actuation residual $B\, r_{\mathrm{slew}}$ is bounded: the applied command (11) and the adaptation target (12) are confined to the pump range $[0, u_{\max}]$ by saturation, and the bounded-gain RLS driven by that saturated target keeps $\hat{\theta}$ — hence the design command (10) — bounded on the compact Ω. (iv) The estimation residual r_{ukf} is Lipschitz in the estimation error e_o, which is uniformly bounded by the regional estimator ISS of Lemma 3 below; that lemma rests only on Assumption 4 and is independent of δ, so citing it here introduces no circularity. Aggregating the four finite bounds gives $\delta \in \mathcal{L}_\infty$. $\square$

Assumption 2 (Reference) *$x_{\mathrm{ref}}, \dot{x}_{\mathrm{ref}}$ are feasible and bounded, so that Ω is compact and forward-invariant under the closed loop.*

Assumption 3 (Bounded gain matrix) *The RLS gain $\Gamma(t)$ is kept uniformly bounded and uniformly positive definite, $0 \prec \gamma_{\min} I \preceq \Gamma(t) \preceq \gamma_{\max} I$, by the forgetting mechanism of Sec. 4.5.2 (with covariance bounding or covariance resetting when persistent excitation is absent).*

Assumption 4 (Estimator regularity) *The joint unscented Kalman filter (UKF) estimator state $x_{\mathrm{UKF}} = [h_{1\ldots4}^T \;\; \gamma_{1,2}^T]^T$ obeys the augmented dynamics $\dot{x}_{\mathrm{UKF}} = F(x_{\mathrm{UKF}}, u) + G_w w + G_\gamma \dot{\gamma}$, where F stacks the level dynamics (2) with the random-walk parameter model ($\dot{\gamma} \equiv 0$ nominally), $G_\gamma\dot{\gamma}$ injects the true valve drift as an exogenous signal, and the measurement is linear in the levels, $y = Cx_{\mathrm{UKF}} + \nu$ with $C = [\,I_4 \;\; 0_{4\times2}\,]$: all four tank levels are sensed, while the valve ratios $\gamma_{1,2}$ are unmeasured, so the sensor block does not act on the γ-block. Let $F_x(t) := \int_0^1 \partial_{x_{\mathrm{UKF}}} F(\hat{x}_{\mathrm{UKF}} + s\, e_o, u)\, ds$ be the averaged state-Jacobian along the estimation error $e_o := x_{\mathrm{UKF}} - \hat{x}_{\mathrm{UKF}}$ and $\Theta(\tau, t)$ the state-transition matrix it generates. On the compact envelope $\Omega \times \mathcal{U}$ (states $\times$ admissible inputs):*

- *(i) (Covariance sandwich) the filter covariance P_o is uniformly bounded and positive definite, $\underline{p} I \preceq P_o(t) \preceq \bar{p} I$ with $\underline{p}, \bar{p} > 0$ (empirically $\underline{p} \approx 4.7 \times 10^{-3}$, $\bar{p} \approx 9.6 \times 10^{-2}$ over the benchmark runs, i.e. $\mathrm{cond}(P_o) \lesssim 20$);*
- *(ii) (Uniform observability) the pair $(F_x(t), C)$ is uniformly completely observable along every admissible trajectory: $\exists T, \alpha, \bar{\alpha} > 0$ with $\alpha I \preceq \int_t^{t+T} \Theta^T(\tau, t)\, C^T R_\nu^{-1} C\, \Theta(\tau, t)\, d\tau \preceq \bar{\alpha} I$, where $R_\nu \succ 0$ is the measurement covariance;*
- *(iii) (Bounded slopes and curvature) $\|F_x(t)\| \le \bar{a}$ and $\sup_\Omega \|\partial^2 F\| \le \bar{H} < \infty$, both finite because F is smooth on the compact Ω (the only nonsmooth can-*

didate, $\partial_h^2\sqrt{h}$, stays bounded since the tanks never drain to $h = 0$);

(iv) (Rate-bounded drift) $\|\dot{\gamma}(t)\| \le \bar{\nu}_\gamma$ for a.e. t, off a finite set of jump instants.

Condition (ii) is the structural crux: although $C = [\,I_4 \;\; 0_{4x2}\,]$ reads all four levels, it cannot see γ directly, so the valve ratios are observable only through the multiplicative input channel $B(\gamma)u$. At the operating point $\bar{h} = [10, 10, 2, 2]^\top$, $\bar{\gamma} = [0.43, 0.34]^\top$ with the steady holding input $\bar{u} \approx [2.6, 1.6]^\top$ V, the discrete observability matrix of (F_x, C) has full column rank 6 (smallest singular value $\sigma_{\min} \approx 3.1 \times 10^{-3}$), but collapses to rank 4 at $u \equiv 0$, losing exactly the two γ-coordinates. Sensing all four levels rather than the (h_1, h_2) pair of Johansson's default keeps this recovery well-conditioned — γ_1 enters both h_1 and h_4, γ_2 both h_2 and h_3, so each ratio is carried by two measured channels. Hence (ii) holds precisely when the inputs persistently excite that channel.

4.3 LTV-MPC Design

Linear Time-Varying MPC (LTV-MPC) re-solves at each control step ($dt = 0.003\,$s), re-linearizing about the current state estimate $\hat{x}_k$, but discretizes its prediction model at the coarser step $\Delta t_p = 0.1\,$s so that the $N = 30$ horizon spans $N\Delta t_p = 3\,$s. This decoupling is deliberate: the tank time constants are on the order of tens of seconds, so a horizon discretized at the 3 ms control step ($N\,dt = 90\,$ms) would be myopic, whereas a 3 s horizon lets the controller anticipate the NMP inverse response. Linearizing (2) about the operating point gives the affine model $\dot{x} = Ax + Bu + d_{\text{lin}}$, where d_{lin} is the linearization offset, the state Jacobian is

$$A = \begin{pmatrix} -\frac{a_1}{A_1}\sqrt{\frac{g}{2h_1}} & 0 & \frac{a_3}{A_1}\sqrt{\frac{g}{2h_3}} & 0 \\ 0 & -\frac{a_2}{A_2}\sqrt{\frac{g}{2h_2}} & 0 & \frac{a_4}{A_2}\sqrt{\frac{g}{2h_4}} \\ 0 & 0 & -\frac{a_3}{A_3}\sqrt{\frac{g}{2h_3}} & 0 \\ 0 & 0 & 0 & -\frac{a_4}{A_4}\sqrt{\frac{g}{2h_4}} \end{pmatrix}, \tag{8}$$

and the input matrix, following Johansson (2000), is

$$B = \begin{pmatrix} \frac{\gamma_1 k_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 k_2}{A_2} \\ 0 & \frac{(1-\gamma_2)k_2}{A_3} \\ \frac{(1-\gamma_1)k_1}{A_4} & 0 \end{pmatrix}, \tag{9}$$

which has full column rank at the NMP operating point (Assumption 1). Cost weighting follows standard state-input scaling guidelines (Mayne et al., 2000; Borrelli et al., 2017), with the QP solved by OSQP (Stellato et al., 2020): prediction horizon $N = 30$, $Q = \text{diag}([40, 40, 5, 5])$, and $R = \text{diag}([0.001, 0.001])$. Horizon, cost weights, and constraints are identical across all runs, with no preemptive tuning.

Algorithm 1 Offline KAN distillation of the LTV-MPC policy

Require: operating set Ω, reference set $\mathcal{R}$, LTV-MPC policy π_{MPC}, KAN architecture
Ensure: frozen basis Φ_{KAN}, initial weights θ^*

```
D ← ∅
for sampled (x, x_ref) ∈ Ω × R do
    ξ ← NORMALIZE(x, x_ref);  u ← π_MPC(x, x_ref)
    D ← D ∪ {(ξ, u)}
end for
train KAN B-spline edges Θ: min_Θ Σ_D ‖KAN_Θ(ξ) − u‖²
Φ_KAN ← SYMBOLICEXTRACT(Θ)    ▷ spline→primitive; compose; simplify
θ* ← linear coefficients of (13)–(14)
freeze Φ_KAN    ▷ only θ adapts online
return Φ_KAN, θ*
```

4.4 Offline KAN Distillation

The regressor Φ_{KAN} of (4) is obtained offline by distilling the LTV-MPC policy of Sec. 4.3 into a fixed symbolic basis (Salkimbayev, 2026). The procedure (Algorithm 1) has three stages. First, the LTV-MPC is rolled out across the operating envelope and reference set, producing a labeled dataset $\mathcal{D} = \{(\xi_i, u_i^{\text{MPC}})\}$ of normalized state/error features ξ and optimal commands. Second, a KAN with learnable B-spline activations on its edges is trained to imitate the policy, $\min_\Theta \sum_i \|\text{KAN}_\Theta(\xi_i) - u_i^{\text{MPC}}\|^2$. Third, each learned univariate spline is replaced by its best-fit symbolic primitive and the network is composed and simplified, yielding the explicit polynomial law (13)–(14): a finite basis $\Phi_{\text{KAN}}(\xi) \in \mathbb{R}^p$ ($p = 10$) with linear coefficients θ^*. The basis is then frozen; only θ adapts online (Sec. 4.5.2). Distillation converts the iterative QP into a constant-time symbolic evaluation while preserving the policy with negligible loss (Salkimbayev, 2026); the present contribution makes this frozen-basis controller adaptive and equips it with the stability certificate of Sec. 4.6. The symbolic law is extracted on normalized inputs and rescaled to physical units at deployment, which ensures benchmark fairness and underlies the regressor bound $\Phi_{\max}$ of Proposition 1.

4.5 Controller Design

4.5.1 Structure of LCA-RLS-KAN

The controller is specified in a design form, in which the error dynamics and the Lyapunov analysis of Sec. 4.6 are conducted, and an implemented form, which the online loop actually evaluates. The design form combines the feedforward/adaptive law with the proportional-integral (PI) law,

$$u_{\text{raw}}(t) = B^{+}\Big(\dot{x}_{\text{ref}} - \Phi_{\text{KAN}}^{T}\hat{\theta} - K\,e(t) - \hat{d}(t)\Big), \quad (10)$$

where B^{+} is the left pseudoinverse of B (Assumption 1; $B \in \mathbb{R}^{4\times 2}$ has full column rank at the NMP operating point), $\dot{x}_{\text{ref}}$ is the feedforward term, $\Phi_{\text{KAN}}^{T}\hat{\theta}$ cancels the learned nonlinearity (4), $K = K_P + \bar{K}_L$ is the total proportional gain (proportional gain K_P plus the Lipschitz-shaping gain $\bar{K}_L := (K_L/e_{\max})I$, see Sec. 4.5.2), and $\hat{d}(t)$ is the integral state with dynamics (20). The sign of the regressor and integral terms in (10) is the cancellation sign required by the error dynamics (26).

The implemented form evaluates (10) composed. Every static term in (10) is affine in the distilled feature dictionary of Sec. 4.4 — B^{+} is a fixed matrix, $\dot{x}_{\text{ref}}$ is constant for the regulation task, and Ke is affine in the level and error features — so the static part of the composed law lies in the span of Φ_{KAN}, and the command applied to the plant is a single saturated dot product per pump,

$$u(t_k) = \text{sat}_{[0,u_{\max}]}\big(u_{\max}[\,\hat{\theta}_1^{T}\varphi_k,\ \hat{\theta}_2^{T}\varphi_k\,]^{T}\big), \quad (11)$$

where $\varphi_k = \Phi_{\text{KAN}}(\xi_k)$ is the regressor evaluated on the normalized features, $u_{\max} = 12$ V is the command denormalization scale, and the saturation confines the command to the physical pump range. The distilled polynomials (13)–(14) are exactly this composition at the initial weights; the composed coefficients are an affine function of the design weights $\hat{\theta}$ (with the integral state entering the bias coordinate), so the two forms carry the same adaptive degrees of freedom. The slowly varying terms of (10) — the integral state $\hat{d}$ and the corrective proportional action — are therefore not added on the actuator line at each instant; they enter through the adaptation channel, carried by the time-varying weights $\hat{\theta}(t)$. At every step a PI-corrected target is formed around the applied command — proportional action on each pump's estimated level error ($K_{\text{corr}} = 3.0$ V/cm, the implementation counterpart of the design gain K), the low-pass-filtered cross-coupled upper-tank error realizing $\hat{d}$ (filter constant $\alpha = 0.05$, weight 0.2), and a saturating cross-coupling brake — and the target, not the actuator command, is rate-limited:

$$\begin{aligned} u_{\text{tgt}}(t_k) = \text{SlewLim}_{\Delta}\big(\tilde{u}_{\text{tgt}}(t_k)\big) &:= u_{\text{tgt}}(t_{k-1}) \\ &+ \text{sat}_{[-\Delta,\Delta]}\big(\tilde{u}_{\text{tgt}}(t_k) - u_{\text{tgt}}(t_{k-1})\big), \end{aligned} \quad (12)$$

with $\tilde{u}_{\text{tgt}}$ the unlimited PI-corrected target: the target may change by at most Δ volts per step ($\Delta = 0.1$ V here) and is saturated to the same $[0, u_{\max}]$ range. The forgetting-RLS of Sec. 4.5.2 folds the residual $u_{\text{tgt}} - u$ into the weights every T_a steps, so the applied command tracks the slew-limited target through the low-gain adaptation rather than through a hard constraint on the actuator line, bounding actuator slew and fatigue while keeping the actuation path branch-free. The realization residual $r_{\text{slew}}(t) := u(t) - u_{\text{raw}}(t)$ between the implemented command (11) and the design command (10) is uniformly bounded — u and u_{tgt} are confined to $[0, u_{\max}]$ by construction, and the bounded-gain RLS driven by this saturated target keeps $\hat{\theta}$, hence u_{raw}, bounded on the compact Ω (Fortescue et al., 1981; Ioannou and Sun, 1996) — and enters the loop additively through $B\,r_{\text{slew}}$, absorbed into the lumped disturbance δ (Lemma 1). The previously monolithic u_L is thus resolved into two well-posed objects: a proportional contribution $-\bar{K}_L e$ folded into K, and the bounded rate saturation (12); this removes the dimensional inconsistency of treating a slew limit as a length.

To demonstrate the transparency of the KAN architecture, the distilled control law for u_1 and u_2 extracts into a single algebraic form (in normalized units; the deployed command is rescaled by $u_{\max}$ and saturated per (11)):

$$\begin{aligned} u_{1,\text{raw}} = &-0.096827\,x_1 - 11.552226\,x_2 + 0.240450\,x_3 \\ &- 1.444357\,x_4 + 3.841384\,x_5 + 0.391351\,x_7 \\ &+ 0.042346\,x_8 + 0.025617\,(-10.000158\,x_1 \\ &- 9.999930)^2 - 0.054467\,(-1.670916\,x_1 \\ &+ 3.952262\,x_3 - 3.957915\,x_4 + 0.059323\,x_8 \\ &- 1.277487)^2 + 1.172892 \end{aligned} \quad (13)$$

$$\begin{aligned} u_{2,\text{raw}} = &\ 0.255780\,x_1 - 2.662247\,x_2 + 0.507210\,x_3 \\ &- 0.526472\,x_4 + 0.555852\,x_5 + 0.094046\,x_7 \\ &- 0.023399\,x_8 + 0.005775\,(-10.000158\,x_1 \\ &- 9.999930)^2 + 0.099910\,(-1.670916\,x_1 + \\ &3.952262\,x_3 - 3.957915\,x_4 + 0.059323\,x_8 \\ &- 1.277487)^2 - 0.470107 \end{aligned} \quad (14)$$

where x_1 is the normalized height of tank 1 (h_1), x_2 is the normalized height of tank 2 (h_2), x_3 is the normalized height of tank 3 (h_3), x_4 is the normalized height of tank 4 (h_4), x_5 is the normalized error of tank 1 ($h_{1target} - h_1$), x_6 is the normalized error of tank 2 ($h_{2target} - h_2$), x_7 is the normalized error of tank 3 ($h_{3target} - h_3$), x_8 is the normalized error of tank 4 ($h_{4target} - h_4$).

The coefficients in (13)–(14) reveal features structurally unavailable in optimization-based controllers: quadratic coefficients encode NMP coupling in the respective pumps and the negative coefficients encode physical damping adjacent to setpoint deviation. The two modes hence provide a multilayered control law, which is specifically designed to handle the inverse response characteristic of the NMP plant configuration.

The modifications ensuring boundedness and ISS are described next.

4.5.2 Lipschitz-Continuous RLS Update Law

On top of the baseline KAN-distilled control law, three modifications enable safe online adaptation.

1) Weight adaptation. The linear weight vector $\hat{\theta}$ is updated by the continuous-time RLS law with forgetting factor λ (Ioannou and Sun, 1996; Slotine and Li, 1991):

$$\dot{\hat{\theta}} = \Gamma \, \Phi \, P \, e, \tag{15}$$

$$\dot{\Gamma} = \rho \, \Gamma - \Gamma \, \Phi \, \Phi^T \Gamma, \qquad \rho \approx \frac{1-\lambda}{T_a}, \tag{16}$$

where $\Phi \equiv \Phi_{\text{KAN}}(x)$ abbreviates the frozen regressor, T_a is the adaptation interval, and $0 < \lambda < 1$. The distilled nonlinear basis is frozen; only the linear weights adapt, so $\tilde{\theta} = \hat{\theta} - \theta^*$ is the weight estimation error (with θ^* constant, $\dot{\tilde{\theta}} = \dot{\hat{\theta}}$) (Fortescue et al., 1981). The discrete realization used in the implementation is the Joseph-stabilized forgetting-RLS recursion; for the per-pump target $y_k = \Phi_k^T \theta + \epsilon_k$,

$$L_k = \Gamma_{k-1} \Phi_k \big(\lambda R_k + \Phi_k^T \Gamma_{k-1} \Phi_k \big)^{-1}, \tag{17}$$

$$\hat{\theta}_k = \hat{\theta}_{k-1} + L_k \big(y_k - \hat{\theta}_{k-1}^T \Phi_k \big), \tag{18}$$

$$\Gamma_k = \tfrac{1}{\lambda} \big[(I - L_k \Phi_k^T) \Gamma_{k-1} (I - L_k \Phi_k^T)^T + L_k R_k L_k^T \big], \tag{19}$$

with R_k being the measurement-noise covariance; the discrete covariance Γ_k is the sampled counterpart of the continuous gain Γ in (16) (with $\rho \approx \frac{1-\lambda}{T_a}$), and the Joseph form (19) keeps it symmetric positive definite. To prevent covariance wind-up when PE is absent, Γ is kept within $[\gamma_{\min} I, \gamma_{\max} I]$ by bounded-gain forgetting / covariance resetting (Assumption 3); constant forgetting alone does not guarantee an upper bound on Γ without excitation.

2) Bounded integral action. The integral state $\hat{d}$ regulates the cumulative deviation from the *setpoint*; its design law is

$$\dot{\hat{d}} = K_I \, e, \qquad K_I = k_I I \succ 0, \tag{20}$$

with the band $[-\bar{d}, \bar{d}]$ kept forward-invariant for $\hat{d}$ (Lemma 2). In implementation this band is not enforced by an active conditional-integration clamp: $\hat{d}$ is realized as the low-pass-filtered error of Sec. 4.5.1 (filter constant α), a leaky accumulation whose output is a convex combination of past errors and is therefore confined to the error range of the compact envelope by construction. The constraint — integration frozen whenever it would push $\hat{d}$ outside $[-\bar{d}, \bar{d}]$ — is the corresponding guardrail for pure-integrator realizations of (20) and is subsumed by the filter structure here. Unlike the weight covariance Γ, the integral gain K_I is constant, so the design update (20) carries no analogue of the forgetting-induced metric decay that damps $\tilde{\theta}$ (Sec. 4.6); boundedness of $\hat{d}$, and hence of $\tilde{d} = \hat{d} - d^*$, is therefore supplied directly by the construction of the update — filter leak or constraint — rather than by the Lyapunov argument. This also removes any need to tie K_I to P: any SPD K_I may be chosen freely for transient shaping.

3) Lipschitz shaping and slew limiting. Lipschitz continuity of the command is enforced through the deviation-dependent proportional gain $\bar{K}_L = (K_L / e_{\max}) I$ (folded into K in (10)) together with the per-step rate saturation (12) of the adaptation target (Khalil and Grizzle, 2002); the applied command (11) is itself Lipschitz in the state, being a saturated polynomial evaluation on the compact Ω (Proposition 1). This keeps the control law restrained and tractable while bounding actuator fatigue.

Lemma 2 (Boundedness of the integral state) *Let the integral update* (20) *be realized so that the band* $[-\bar{d}, \bar{d}]$ *is forward-invariant for* $\hat{d}$ *— by the leaky low-pass realization implemented here, whose output is a convex combination of past errors, or by a conditional-integration (anti-windup) clamp. Then* $\|\hat{d}(t)\| \le \bar{d}$ *for an explicit, known* $\bar{d}$, $\forall t$, *by construction of the update rather than as a consequence of the Lyapunov argument of Sec. 4.6. Together with* $\|d^*\| \le d_{\max}$ *(Lemma 1), the integral-state error obeys the uniform bound* $\|\tilde{d}(t)\| \le \bar{d} + d_{\max} =: \bar{d}_{\text{tot}}$ $\forall t$.

Proof. Under the implemented leaky realization the discrete update is the convex combination $\hat{d}^+ = (1-\alpha)\hat{d} + \alpha \, e$: with $\hat{d}(0) = 0$, coordinatewise $|\hat{d}_i(t)| \le \max_{\tau \le t} |e_i(\tau)|$, and since the error is confined to the compact envelope Ω (Assumption 2), any $\bar{d} \ge \sup_\Omega \|e\|_\infty$ renders the band positively invariant a priori. Under a clamp, integration acts coordinatewise on the accumulation (20): whenever an increment $K_I e \, \Delta t$ would carry $\hat{d}$ outside $[-\bar{d}, \bar{d}]$, it is suppressed and $\hat{d}$ is held at the boundary. In either case $\|\hat{d}(t)\| \le \bar{d}$ for all t, independently of any Lyapunov estimate. With $\tilde{d} = \hat{d} - d^*$ and $\|d^*\| \le d_{\max}$ (Lemma 1), the triangle inequality gives $\|\tilde{d}\| \le \bar{d} + d_{\max} =: \bar{d}_{\text{tot}}$. Empirically the tighter band $\bar{d} = 4.0$ cm is certified by the logs: across the Monte-Carlo campaign of Sec. 5 the integral coordinate never exceeds 3.87 cm. □

The complete online cycle — estimation, constant-time inference, saturated actuation with rate-limited target shaping, and periodic weight adaptation — is summarized in Algorithm 2.

Algorithm 2 Online LCA-RLS-KAN control step at time t_k

Require: frozen Φ_{KAN}, scale $u_{\max}$, weights $\hat{\theta}_1, \hat{\theta}_2$, covariances Γ_1, Γ_2, forgetting λ, slew Δ, interval T_a, UKF

UKF predict with $u(t_{k-1})$; update with $z_k \Rightarrow \hat{x}_k = [\hat{h}, \hat{\gamma}]$

$\varphi \leftarrow \Phi_{\mathrm{KAN}}(\hat{x}_k, x_{\mathrm{ref}})$ $\triangleright$ $\mathcal{O}(p)$, branch-free

$u_{\mathrm{raw}} \leftarrow u_{\max}[\,\hat{\theta}_1^\top \varphi,\ \hat{\theta}_2^\top \varphi\,]$ $\triangleright$ constant-time inference

$u(t_k) \leftarrow \mathrm{sat}_{[0,u_{\max}]}\big(u_{\mathrm{raw}}(t_k)\big)$ (11); apply to plant

form PI-corrected target $\tilde{u}_{\mathrm{tgt}}$ (filtered error + cross-coupling brake); $u_{\mathrm{tgt}} \leftarrow \mathrm{SlewLim}_\Delta\big(\tilde{u}_{\mathrm{tgt}}\big)$ (12)

if $k \bmod T_a = 0$ **then**

  for $j \in \{1, 2\}$ **do**

    $(\hat{\theta}_j, \Gamma_j) \leftarrow \text{FF-RLS}(\hat{\theta}_j, \Gamma_j, \varphi,\ u_{\mathrm{tgt},j} - u_j; \lambda)$

  end for

end if

4.5.3 Lyapunov Candidate

The energy function is taken over the tracking and weight errors only,

$$V(\zeta) = \tfrac{1}{2} e^T P e + \tfrac{1}{2} \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}, \tag{21}$$

with $\zeta = [\,e^T\ \tilde{\theta}^T\,]^T$, $e = x - x_{\mathrm{ref}}$ and $\tilde{\theta}$ the weight estimation error. The integral-state error $\tilde{d} = \hat{d} - d^*$ is deliberately excluded from V: by Lemma 2 it is already uniformly bounded by construction of its band-confined realization, independently of any Lyapunov argument, and it is instead carried into the analysis below as a bounded perturbation acting on the $(e, \tilde{\theta})$-subsystem. Since P and Γ^{-1} (Assumption 3) are symmetric positive definite, V is positive definite and radially unbounded, and there exist $c_1, c_2 > 0$ with

$$c_1 \|\zeta\|^2 \le V(\zeta) \le c_2 \|\zeta\|^2, \tag{22}$$

i.e. V satisfies class-$\mathcal{K}_\infty$ bounds on the same vector ζ that the dissipation rate of Sec. 4.6 will control (cf. Khalil and Grizzle (2002, Lemma 4.3)).

4.6 Stability Analysis

Differentiating (21) along the closed-loop trajectories,

$$\dot{V} = e^T P \dot{e} + \tilde{\theta}^T \Gamma^{-1} \dot{\tilde{\theta}} + \tfrac{1}{2} \tilde{\theta}^T \tfrac{d}{dt}\big(\Gamma^{-1}\big) \tilde{\theta}. \tag{23}$$

With the plant model (4), split the approximation error into its constant and time-varying parts, $\varepsilon = d^* + \varepsilon_{\mathrm{tv}}(x)$, with d^* being the constant bias. Substituting the command (10)–(12) into $\dot{e}$, we get:

$$\dot{e} = \dot{x} - \dot{x}_{\mathrm{ref}} = f(x) + Bu - \dot{x}_{\mathrm{ref}}; \tag{24}$$

using $BB^+ \approx I$, collecting the time-varying terms into the lumped disturbance δ:

$$\delta = \varepsilon_{\mathrm{tv}} + r_{\mathrm{ua}} + B\, r_{\mathrm{slew}} + r_{\mathrm{ukf}} \tag{25}$$

of Lemma 1 (so the approximation-error remainder $\varepsilon - d^* = \varepsilon_{\mathrm{tv}}$ is only one of its four components), and setting $\theta^* - \hat{\theta} = -\tilde{\theta}$, $d^* - \hat{d} = -\tilde{d}$, the error dynamics are

$$\dot{e} = -\Phi^T \tilde{\theta} - \tilde{d} - Ke + \delta(x), \qquad K = K_P + \bar{K}_L. \tag{26}$$

Group the last two exogenous terms into a single effective disturbance:

$$\omega(t) := \delta(x(t), t) - \tilde{d}(t), \tag{27}$$

so that $\dot{e} = -\Phi^T \tilde{\theta} - Ke + \omega$. Because $\tilde{d}$ is bounded by construction (Lemma 2) rather than by anything established below, ω may be treated as an exogenous bounded signal acting on the $(e, \tilde{\theta})$-subsystem: $\|\omega(t)\| \le \|\delta\|_\infty + \bar{d}_{\mathrm{tot}} =: \bar{\omega} < \infty$, uniformly in t. Since θ^* is constant, $\dot{\tilde{\theta}} = \dot{\hat{\theta}}$ is given by (15). The gain law (16) yields the inverse-covariance derivative

$$\tfrac{d}{dt}\big(\Gamma^{-1}\big) = \Phi\Phi^T - \rho\, \Gamma^{-1}. \tag{28}$$

Substituting (26)–(28) into (23), the weight cross-term $\tilde{\theta}^T \Gamma^{-1}(\Gamma \Phi P e) = \tilde{\theta}^T \Phi P e$ cancels $-e^T P \Phi^T \tilde{\theta}$ exactly (both equal $e^T P \Phi^T \tilde{\theta}$ by symmetry of P), leaving

$$\dot{V} = -e^T P K e + e^T P \omega + \tfrac{1}{2} \tilde{\theta}^T \Phi \Phi^T \tilde{\theta} - \tfrac{1}{2} \rho\, \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}. \tag{29}$$

No cross-term involving $\tilde{d}$ remains to be cancelled or bounded here, because $\tilde{d}$ was never placed in V in the first place (Sec. 4.5.3); its effect on the loop enters only through the bounded forcing term ω in (29), exactly as δ does. The cross/indefinite terms are bounded by Cauchy–Schwarz and Young's inequalities. For the disturbance term, with $\mu > 0$,

$$e^T P \omega \le \lambda_{\max}(P)\, \|e\|\, \|\omega\| \le \tfrac{\mu}{2} \|e\|^2 + \tfrac{\lambda_{\max}(P)^2}{2\mu} \|\omega\|^2. \tag{30}$$

By Proposition 1, $\frac{1}{2} \tilde{\theta}^T \Phi \Phi^T \tilde{\theta} \le \frac{1}{2} \Phi_{\max}^2 \|\tilde{\theta}\|^2$, and combining with the forgetting term,

$$\begin{aligned} &\tfrac{1}{2} \Phi_{\max}^2 \|\tilde{\theta}\|^2 - \tfrac{1}{2} \rho\, \tilde{\theta}^T \Gamma^{-1} \tilde{\theta} \le \\ &-\tfrac{1}{2} \big(\rho\, \lambda_{\min}(\Gamma^{-1}) - \Phi_{\max}^2\big) \|\tilde{\theta}\|^2. \end{aligned} \tag{31}$$

This term is exactly the "implicit leakage" of exponential forgetting: Γ^{-1} decaying at rate ρ contributes a genuine negative-definite term to $\dot{V}$ through the metric itself (not through any explicit pull-to-zero in $\dot{\hat{\theta}}$), provided Γ^{-1} stays bounded away from 0 (Assumption 3).

The design integrator (20) has no such mechanism — K_I is constant, so no analogous term arises for $\tilde{d}$; this is precisely why $\tilde{d}$ is handled by construction (Lemma 2) rather than by the Lyapunov dissipation; a leak placed on $\hat{d}$ itself is what the companion σ-modified design of Sec. 4.7 would convert into dissipation.

Taking the gains scalar, $K = kI$ with $k = k_P + \bar{k}_L > 0$, so $PK = kP \succ 0$ and $\lambda_{\min}(PK) = k\,\lambda_{\min}(P)$ (more generally, it suffices that $PK + K^T P \succ 0$). Requiring the error term to dominate the disturbance cross-term, $\lambda_{\min}(PK) > \mu/2$; choose $\mu \in (0, 2\lambda_{\min}(PK))$, set $\mu = \lambda_{\min}(PK)$. By Assumption 3, Γ^{-1} is bounded with $\lambda_{\min}(\Gamma^{-1}) > 0$, and the damping condition

$$\rho\,\lambda_{\min}(\Gamma^{-1}) > \Phi_{\max}^2 \tag{32}$$

is enforceable since $\Phi_{\max}$ is finite (Proposition 1). Collecting (30)–(31), the time derivative satisfies

$$\begin{aligned}\dot{V} &\le -\Big(\lambda_{\min}(PK) - \tfrac{\mu}{2}\Big)\|e\|^2 \\ &-\tfrac{1}{2}\Big(\rho\,\lambda_{\min}(\Gamma^{-1}) - \Phi_{\max}^2\Big)\|\tilde{\theta}\|^2 + \tfrac{\lambda_{\max}(P)^2}{2\mu}\|\omega\|^2.\end{aligned} \tag{33}$$

This is a negative-definite term in every component of $\zeta = (e, \tilde{\theta})$, with a forcing term bounded uniformly by $\bar{\omega}$ (Lemma 2); the integral channel contributes only through this bounded forcing, never through an unbounded or unaccounted state.

Define

$$\begin{aligned}\alpha &:= \min\Big\{\lambda_{\min}(PK) - \tfrac{\mu}{2},\ \tfrac{1}{2}\big(\rho\lambda_{\min}(\Gamma^{-1}) - \Phi_{\max}^2\big)\Big\}, \\ \gamma &:= \tfrac{\lambda_{\max}(P)^2}{2\mu},\end{aligned} \tag{34}$$

both positive under (32). Then (33) reads $\dot{V} \le -\alpha\|\zeta\|^2 + \gamma\|\omega\|^2$, a dissipation inequality on the same state $\zeta = [\,e^T\ \tilde{\theta}^T\,]^T$ bounded in (22), with ω as in (27) satisfying $\|\omega\|_\infty \le \bar{\omega}$. The UKF estimation residual is treated as a component of δ (Lemma 1) and handled by an interconnection (cascade-ISS) argument rather than a separation principle: the estimator subsystem is shown regionally ISS in Lemma 3, and its interconnection with the controller subsystem is closed in Lemma 4.

Theorem 1 (Stability of LCA-RLS-KAN)
Consider the closed-loop system formed by the plant (1) *and the LCA-RLS-KAN controller* (10)–(12), (15)–(20), *under Proposition 1, Assumptions 1–3, and Lemmas 1 and 2. Let the gains P, Γ be SPD, $K = kI \succ 0$, and let $\hat{d}$ be realized so that the band invariance of Lemma 2 holds (leaky low-pass realization or anti-windup clamp). If the damping condition* (32) *holds, then with α, γ in* (34) *and ω as in* (27),

$$\dot{V} \le -\alpha\,\|\zeta\|^2 + \gamma\,\|\omega\|^2. \tag{35}$$

By (22), *V is an ISS-Lyapunov function in the sense of Sontag (1989) (cf. Khalil and Grizzle (2002, Def. 4.7, Thm. 4.19)); hence the $(e, \tilde{\theta})$-subsystem is input-to-state stable with respect to ω, and $\zeta = (e, \tilde{\theta})$ is globally uniformly ultimately bounded, converging to the residual set*

$$\mathcal{R} = \Big\{\zeta :\ \|\zeta\| \le \sqrt{\tfrac{c_2}{c_1}\tfrac{\gamma}{\alpha}}\,\bar{\omega}\Big\}. \tag{36}$$

The integral-state error $\tilde{d}$ is bounded independently by $\bar{d}_{\text{tot}}$ (Lemma 2), so the full triple $(e, \tilde{\theta}, \hat{d})$ remains bounded $\forall t$.

Proof. The dissipation inequality (35) was assembled in (23)–(34); it remains to convert it into an explicit bound. By the lower bound in (22), $\|\zeta\|^2 \ge V/c_2$, so (35) with $\|\omega\|_\infty \le \bar{\omega}$ gives the scalar comparison inequality

$$\dot{V} \le -\eta\,V + \gamma\bar{\omega}^2, \qquad \eta := \frac{\alpha}{c_2} > 0. \tag{37}$$

The comparison lemma (Khalil and Grizzle, 2002, Lemma 3.4) integrates (37) to

$$V(t) \le e^{-\eta t}V(0) + \frac{\gamma}{\eta}\big(1 - e^{-\eta t}\big)\bar{\omega}^2 \le e^{-\eta t}V(0) + \frac{c_2\gamma}{\alpha}\bar{\omega}^2. \tag{38}$$

Applying $c_1\|\zeta\|^2 \le V(t)$ and $V(0) \le c_2\|\zeta(0)\|^2$ from (22) together with $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$,

$$\|\zeta(t)\| \le \underbrace{\sqrt{\tfrac{c_2}{c_1}}\,\|\zeta(0)\|\,e^{-\eta t/2}}_{\beta(\|\zeta(0)\|,\,t)\,\in\,\mathcal{KL}} + \underbrace{\sqrt{\tfrac{c_2}{c_1}\tfrac{\gamma}{\alpha}}\,\bar{\omega}}_{\kappa(\|\omega\|_\infty)\,\in\,\mathcal{K}}. \tag{39}$$

Estimate (39) is exactly an ISS bound — a class-$\mathcal{KL}$ transient in $(\|\zeta(0)\|, t)$ plus a class-$\mathcal{K}$ function of $\|\omega\|_\infty$ (Sontag, 1989), (Khalil and Grizzle, 2002, Thm. 4.19). Hence the $(e, \tilde{\theta})$-subsystem is ISS with respect to ω, ζ is GUUB, and

$$\limsup_{t\to\infty}\|\zeta(t)\| \le \sqrt{\tfrac{c_2}{c_1}\tfrac{\gamma}{\alpha}}\,\bar{\omega}, \tag{40}$$

i.e. ζ enters the residual set (36), approached at exponential rate $\eta/2 = \alpha/(2c_2)$. The integral coordinate is bounded separately, $\|\tilde{d}(t)\| \le \bar{d}_{\text{tot}}$ for all t by Lemma 2, so $(e, \tilde{\theta}, \hat{d})$ is bounded while $(e, \tilde{\theta})$ is GUUB. □

Lemma 3 (Regional ISS of the joint estimator)
Under Assumption 4, realize the UKF as the output-injection observer

$$\begin{aligned}\dot{\hat{x}}_{\text{UKF}} &= F(\hat{x}_{\text{UKF}}, u) + K_{\text{f}}(t)\,(y - C\hat{x}_{\text{UKF}}), \\ K_{\text{f}} &= P_o C^T R_\nu^{-1}.\end{aligned} \tag{41}$$

Then the estimation error $e_o = x_{\rm UKF} - \hat{x}_{\rm UKF}$ is regionally ISS with respect to $(\nu, \dot\gamma)$, with gains uniform over $\Omega\times\mathcal{U}$:

$$\|e_o(t)\| \le \beta_o(\|e_o(0)\|, t) + \rho_o(\|(\nu,\dot\gamma)\|_\infty) + b_o, \qquad (42)$$

for some $\beta_o \in \mathcal{KL}$, $\rho_o \in \mathcal{K}$, and a bias floor $b_o = \mathcal{O}(\bar{H}\bar{p})$ induced by unscented statistical linearization (vanishing as the filter is tightened, $\bar p\downarrow$, or the plant curvature is mild, $\bar H\downarrow$).

Proof. Subtracting (41) from the augmented plant and using the mean-value form $F(x_{\rm UKF}, u) - F(\hat{x}_{\rm UKF}, u) = F_x(t)\, e_o$ (exact, with F_x as in Assumption 4) gives

$$\dot e_o = \big(F_x - K_{\rm f} C\big) e_o + \Delta F_x\, e_o - K_{\rm f}\nu + G_w w + G_\gamma \dot\gamma + b_{\rm UT}, \qquad (43)$$

where $\Delta F_x := \hat F_x - F_x$ is the gap between the sigma-point (unscented) slope $\hat F_x$ actually used by the filter and the true averaged Jacobian, and $b_{\rm UT}$ is the unscented mean bias. By Assumption 4(i),(iii) both are bounded on Ω: $\|\Delta F_x\| \le \kappa_{\rm UT} = \mathcal{O}(\bar H\sqrt{\bar p})$ and $\|b_{\rm UT}\| \le \frac{1}{2}\bar H \,\mathrm{tr}\, P_o \le \frac12 n\bar H\bar p$.

Take $V_o = e_o^T P_o^{-1} e_o$, so $\bar p^{-1}\|e_o\|^2 \le V_o \le \underline{p}^{-1}\|e_o\|^2$ by (i). The filter Riccati equation $\dot P_o = F_x P_o + P_o F_x^T + G_w Q_w G_w^T - P_o C^T R_\nu^{-1} C P_o$ yields, for the nominal matrix $\bar F := F_x - K_{\rm f} C$,

$$\begin{aligned}&\bar F^T P_o^{-1} + P_o^{-1}\bar F + \tfrac{d}{dt}P_o^{-1}\\ &= -\big(C^T R_\nu^{-1} C + P_o^{-1} G_w Q_w G_w^T P_o^{-1}\big) \preceq 0.\end{aligned} \qquad (44)$$

By the covariance sandwich (i) and uniform observability/controllability (ii), the integral of the right-hand side over the window T is coercive, so $\bar F(t)$ is uniformly exponentially stable and the nominal flow satisfies $\dot V_o\big|_{\rm nom} \le -c_o V_o$ for some $c_o > 0$ (Kalman–Bucy observer stability (Jazwinski, 1970)). The perturbation $\Delta F_x e_o$ is dominated by this margin whenever $\kappa_{\rm UT}/\underline{p}$ is a small enough fraction of c_o; bounding the remaining terms with Young's inequality and using that $K_{\rm f} = P_o C^T R_\nu^{-1}$ is bounded (by (i), $R_\nu \succ 0$) gives, for some $\epsilon \in (0, c_o)$,

$$\dot V_o \le -(c_o-\epsilon)V_o + \kappa_1\|\nu\|^2 + \kappa_2\|\dot\gamma\|^2 + \kappa_3\|w\|^2 + \kappa_4\|b_{\rm UT}\|^2, \qquad (45)$$

with every $\kappa_i < \infty$. The comparison lemma and the V_o-sandwich convert (45) into the bound (42), with the constant term collecting the $b_{\rm UT}$ contribution as $b_o = \mathcal{O}(\bar H\bar p)$ and process noise w folded into the input vector. The estimate is regional because (iii) holds on Ω, and uniform over $\Omega\times\mathcal{U}$ because (ii) holds along every admissible trajectory, so β_o, ρ_o, b_o do not depend on the controller substate. □

Remark. The augmented model is affine in the unknown parameters: collecting the square-root outflows of (2) in $\varphi(h)$ and the known input channels in $\Psi(u)$, the level dynamics read $\dot h = \varphi(h) + \Psi(u)\,\gamma$ — a state-affine system with γ entering linearly. This is exactly the class for which adaptive-observer theory (Marino and Tomei, 1992; Besançon, 2000; Zhang, 2002) guarantees exponentially convergent joint state/parameter estimation that is ISS to measurement noise and to the parameter-variation rate $\dot\gamma$, under a persistency-of-excitation condition on the regressor $\Psi(u)$. That PE requirement is precisely the structural content of Assumption 4(ii) — the valve ratios are unobservable from the level sensors at $u\equiv 0$ and become uniformly observable when the inputs excite the multiplicative channel — and the same excitation it demands is the regime whose absence is quantified controller-side in Corollary 2. The drift enters Lemma 3 as the standard time-varying-parameter input: the constant-rate valve ramp ($\dot\gamma_1 = -0.0011$/s) is a bounded $\|\dot\gamma\|_\infty$ absorbed by ρ_o, whereas the mid-run 20% step in γ_2 is an impulsive $\dot\gamma$ excluded from (iv) and instead handled as a bounded jump in e_o that re-engages the $\beta_o(\cdot, t)$ transient rather than the $\mathcal{K}$ gain — the ramp/step split mapping cleanly onto the $\mathcal{K}/\mathcal{KL}$ decomposition of ISS.

Lemma 4 (Observer-controller feedback) *Let the UKF residual entering Lemma 1 be Lipschitz in the estimation error, $\|r_{\rm ukf}\| \le L_o\|e_o\|$ on Ω. Under Assumption 4, the estimator bound of Lemma 3 holds with gains uniform over $\Omega\times\mathcal{U}$, and the closed loop keeps (x, u) in that compact envelope (Assumption 2). Then the interconnection of the estimator with the ISS controller subsystem of Theorem 1 is ISS, and $(e, \tilde\theta)$ remains GUUB.*

Proof. The controller substate enters the estimator only through $u = u(\hat x, \hat\theta, \hat d)$, so the interconnection is genuine feedback, not a one-way cascade; two facts collapse it to a cascade in the ISS sense. First, by Lemma 3 the estimator gains β_o, ρ_o, b_o are uniform over $\Omega\times\mathcal{U}$ and so do not depend on $(e, \tilde\theta)$ beyond the requirement that (x, u) remain in the compact envelope, which Assumption 2 guarantees — and which is physical rather than circular here: the tank levels are confined to $[0, h_{\max}]$ by finite capacity and the commands are saturated to $[0, u_{\max}] = [0, 12]$ V by (11), so (x, u) lies in a fixed compact set independently of the stability argument. Hence e_o obeys an exogenous-input bound independent of the controller state. Second, by Theorem 1 the controller subsystem is ISS with respect to ω, and ω depends on the estimator only through $r_{\rm ukf}$ with $\|r_{\rm ukf}\| \le L_o\|e_o\|$. Substituting the uniform estimator bound renders the controller input a class-$\mathcal{K}$ function of $\|(\nu,\dot\gamma)\|_\infty$ plus a class-$\mathcal{KL}$ transient and the constant $L_o b_o$; since a cascade of ISS systems with uniform gains is ISS (Sontag, 1989; Khalil and Grizzle, 2002), the interconnection is ISS and (39) holds with $\bar\omega$ replaced by the composite gain $\bar\omega + L_o\rho_o(\|(\nu,\dot\gamma)\|_\infty) + L_o b_o$. The estimator-ISS premise is now established (Lemma 3) under the verifiable conditions of Assumption 4 rather than assumed; filter consistency (NIS $\approx$ 3.995) and the parameter bounds of

Table 5 corroborate the covariance sandwich (i) and the bias floor b_o empirically. □

Corollary 1 (Practical tracking accuracy) *Consider the idealized regime in which the entire lumped disturbance vanishes, $\delta \equiv 0$ — i.e. the distilled basis captures the nonlinearity exactly ($\varepsilon_{\rm tv} \equiv 0$), the input map is invertible on the operating subspace ($r_{\rm ua} \equiv 0$), the implemented command realizes the design law exactly ($r_{\rm slew} \equiv 0$), and the estimator residual is negligible ($r_{\rm ukf} \equiv 0$) — with no constant bias ($d^* = 0$). The tracking error asymptotically satisfies*

$$\|e(t)\| \lesssim \sqrt{\tfrac{c_2}{c_1}\,\tfrac{\gamma}{\alpha}}\;\bar{d}_{\rm tot} \qquad \textit{as } t \to \infty. \tag{46}$$

Exact asymptotic tracking ($\lim_{t\to\infty} e(t) = 0$) is not claimed here: the band-confined integral state (Sec. 4.5.2) need not drive $\tilde{d}$ to zero even when $\delta \equiv 0$ — it merely stays constrained. The corollary instead makes explicit that achievable tracking accuracy in the ideal case is a designer-controllable quantity, set directly by the integral band $\bar{d}$: tightening the band — through the filter realizing $\hat{d}$ or an explicit constraint — tightens the guaranteed accuracy, at the usual cost of a less aggressive integrator.

Proof. Let $\delta \equiv 0$ — i.e. every component of the lumped disturbance vanishes ($\varepsilon_{\rm tv} \equiv 0$, $r_{\rm ua} \equiv 0$, $r_{\rm slew} \equiv 0$, $r_{\rm ukf} \equiv 0$) — with no constant bias ($d^* = 0$). Then by (27), $\omega = -\tilde{d}$, so $\|\omega\|_\infty \le \bar{d}_{\rm tot}$ by Lemma 2 alone — the residual disturbance entering the proof is then governed entirely by the tightness of the integral band. By (36), we get (46). □

Corollary 2 (Robustness to lack of excitation) *Under the conditions of Theorem 1, the error state remains GUUB even without persistent excitation.*

Proof. Assumption 3 supplies the covariance-bounding (resetting) mechanism that enforces $\gamma_{\min} I \preceq \Gamma(t) \preceq \gamma_{\max} I$, so Γ^{-1} stays bounded and (32) continues to hold; consequently the bound (33) is unaffected by the loss of excitation. The weight error $\tilde{\theta}$ then merely fails to converge to zero but stays in a bounded set, and ISS of (35) preserves GUUB of ζ (Ioannou and Sun, 1996). This is consistent with the persistent-excitation premise of Lemma 3, which is required only to render the estimation error e_o convergent: even in absence of excitation, the covariance sandwich (Assumption 4(i)) still keeps e_o uniformly bounded, and it is only this boundedness — entering the controller through the residual $r_{\rm ukf}$ — that the present GUUB conclusion invokes. □

As such, the Lyapunov-based proof is concluded, providing a verification of the robustness claim made earlier.

4.7 Companion Design: σ-Modified Integrator

Theorem 1 keeps the integral coordinate outside the Lyapunov certificate, bounding it by construction (Lemma 2). We now show that a single leakage term brings $\tilde{d}$ inside the guarantee, certifying the full composite state $z = [\,e^T\ \tilde{\theta}^T\ \tilde{d}^T\,]^T$. Replace the integral update (20) with the σ-modified law

$$\dot{\hat{d}} = K_I\, e - \sigma_I \hat{d}, \qquad \sigma_I > 0,\ \ K_I = k_I I \succ 0, \tag{47}$$

and augment the candidate (21) with an integral metric,

$$V_z(z) = \tfrac{1}{2} e^T P e + \tfrac{1}{2}\tilde{\theta}^T \Gamma^{-1}\tilde{\theta} + \tfrac{1}{2}\tilde{d}^T Q_I \tilde{d}, \quad Q_I = Q_I^T \succ 0. \tag{48}$$

The choice $Q_I = P/k_I$ gives $Q_I K_I = P$, so the integral cross-term cancels the $-e^T P \tilde{d}$ contribution of $e^T P \dot{e}$ exactly. With $\tilde{d} = \hat{d} - d^*$ and constant d^*,

$$\tilde{d}^T Q_I \dot{\tilde{d}} = \underbrace{\tilde{d}^T P e}_{\text{cancels}} - \sigma_I \tilde{d}^T Q_I \tilde{d} - \sigma_I \tilde{d}^T Q_I d^*, \tag{49}$$

and Young's inequality on the last term, $-\sigma_I \tilde{d}^T Q_I d^* \le \frac{\sigma_I}{2}\|\tilde{d}\|^2_{Q_I} + \frac{\sigma_I}{2}\|d^*\|^2_{Q_I}$, leaves the negative-definite integral term $-\frac{\sigma_I}{2}\tilde{d}^T Q_I \tilde{d}$. Adjoining this to (33),

$$\begin{aligned}\dot{V}_z \le &-\big(\lambda_{\min}(PK) - \tfrac{\mu}{2}\big)\|e\|^2 \\ &- \tfrac{1}{2}\big(\rho\lambda_{\min}(\Gamma^{-1}) - \Phi^2_{\max}\big)\|\tilde{\theta}\|^2 - \tfrac{\sigma_I}{2}\lambda_{\min}(Q_I)\|\tilde{d}\|^2 \\ &+ \tfrac{\lambda_{\max}(P)^2}{2\mu}\|\delta\|^2 + \tfrac{\sigma_I}{2}\|d^*\|^2_{Q_I}.\end{aligned} \tag{50}$$

Theorem 2 (Full-state ISS of σ-modification) *Consider the closed loop of Sec. 4.5.1 with* (20) *replaced by* (47) *and V_z as in* (48) *with $Q_I = P/k_I$. Under Proposition 1, Assumptions 1–3, Lemma 1, and the damping condition* (32)*, define*

$$\begin{aligned}\alpha_z := \min\Big\{&\lambda_{\min}(PK) - \tfrac{\mu}{2}, \\ &\tfrac{1}{2}\big(\rho\lambda_{\min}(\Gamma^{-1}) - \Phi^2_{\max}\big),\ \tfrac{\sigma_I}{2}\lambda_{\min}(Q_I)\Big\} > 0\end{aligned} \tag{51}$$

and $\varpi := (\delta, d^*)$. Then $\dot{V}_z \le -\alpha_z\|z\|^2 + \gamma_z\|\varpi\|^2$, where γ_z collects the two forcing constants of (50)*. Hence $z = [\,e^T\ \tilde{\theta}^T\ \tilde{d}^T\,]^T$ — now including the integral coordinate — is ISS with respect to ϖ and GUUB, converging to a residual set of radius $\sqrt{(c_2'/c_1')(\gamma_z/\alpha_z)}\,\|\varpi\|_\infty$, with c_1', c_2' the class-$\mathcal{K}_\infty$ bound constants of V_z.*

Proof. Equation (50) is a dissipation inequality of the same form as (35) with $(z, V_z, \alpha_z, \gamma_z)$ in place of $(\zeta, V, \alpha, \gamma)$; the comparison-lemma argument (37)–(40) then applies verbatim. □

The σ_I leak trades exact integral action for a bounded steady-state offset of order σ_I (the constant $\frac{\sigma_I}{2}\|d^*\|^2_{Q_I}$), the standard σ-modification trade-off (Ioannou and Sun, 1996). Theorem 1 (constrained) and Theorem 2 (σ-modified) are thus two consistent robustifications of the same controller: the former bounds $\tilde{d}$ by construction, the latter by dissipation. The implemented low-pass error filter of Sec. 4.5.2 is not a realization of (47), and the two act at different points of the loop: σ-modification intervenes downstream, leaking an already-formed estimate toward zero — classically $\hat{\theta}$ (Ioannou and Sun, 1996), here the bias estimate $\hat{d}$ — whereas the implemented filter intervenes upstream, contracting the raw error to a convex combination and hence bounding it before it reaches the integral action or the adaptation target. It shrinks accumulated error, never a parameter estimate. Its guarantee therefore runs through Lemma 2 and Theorem 1; (47) is the companion design whose certificate extends to the integral coordinate itself.

5 Results & Discussion

Simulations were conducted in Python 3.12.4 over a 50 s horizon ($dt = 3\,\mathrm{ms}$, $N_{\mathrm{sim}} = 16666$ steps) on a Ryzen 7 7840HS laptop. A joint UKF (Kalman, 1960; Wan and Van Der Merwe, 2000; Julier and Uhlmann, 2004) (FilterPy (Labbe, 2014)) estimates the augmented state $x_{\mathrm{UKF}} = [h_{1\ldots4}, \gamma_{1,2}]$ and is tuned for nominal consistency (NIS $\approx$ 3.995 across all runs). The same estimator and simulation harness are shared by every controller, so reported differences are attributable to the control law alone.

Baselines. Three baselines are compared against LCA-RLS-KAN. (i) An **RBF-NN** adaptive controller sharing the identical adaptation harness (joint UKF, forgetting-RLS weight update, slew-limited target) but replacing the distilled symbolic regressor with a black-box Gaussian radial-basis dictionary; this isolates the effect of the regressor choice. (ii)–(iii) An **LTV-MPC** ($N = 30$, $Q = \mathrm{diag}(40, 40, 5, 5)$, $R = 10^{-3}I$, OSQP (Stellato et al., 2020)) in two configurations: warm-started, reusing the previous solution, and cold-restarted, resolving from scratch each step as in a naive per-step redeployment. Both MPC configurations are reported deliberately: the actuator smoothness of MPC turns out to be implementation-dependent, and the contrast is itself diagnostic (Sec. 5.2).

Monte-Carlo protocol. Performance is evaluated over a **50-seed Monte-Carlo**. Each seed draws fresh zero-mean process ($\sigma_0^2 = 10^{-4}$) and measurement ($\sigma_{\mathrm{m}}^2 = 2.37$) noise sequences, applied under an identical adversarial schedule (valve γ_1 drift -0.0011/s; γ_2 step -20% at $t = 25\,\mathrm{s}$). All four controllers are run on every seed; we report the median and interquartile range (IQR) over the 50 realizations. This allows us to conduct a genuine sampling study over independent draws.

Table 3
Monte-Carlo tracking error, IAE [cm·s], median [IQR] over 50 seeds (lower is better).

Controller	Tank 1	Tank 2	Tank 3	Tank 4
LCA-RLS-KAN	50.9 [1.6]	41.7 [1.0]	38.2 [1.5]	88.5 [2.1]
RBF-NN	50.7 [1.5]	40.9 [0.9]	38.0 [1.2]	**88.3** [2.1]
LTV-MPC (warm)	**30.5** [1.0]	39.6 [1.1]	**34.2** [1.1]	114.9 [4.7]
LTV-MPC (cold)	31.1 [0.9]	**39.3** [1.1]	34.4 [1.1]	114.5 [4.5]

5.1 Tracking and Cross-Coupled Actuation

Table 3 and Fig. 2 report the per-tank IAE distributions. The directly actuated setpoints (tanks 1–3) are tracked most tightly by LTV-MPC, as expected of a controller that explicitly minimizes a quadratic tracking cost on those states. The decisive metric, however, is tank 4: the only level held below its open-loop equilibrium, reachable solely through the cross-coupled $(1 - \gamma_1)$ pathway of pump 1. Here both MPC configurations let tank 4 deviate further (IAE $\approx$ 115 cm·s), approximately 30% worse than LCA-RLS-KAN and RBF-NN ($\approx$ 88 cm·s); crucially, this gap persists under the 3 s prediction horizon (Sec. 4.3), so it reflects the controller's cost/actuation trade-off rather than a short-sighted horizon. MPC purchases tight tracking of the easy, directly actuated states by improperly engaging the coupled actuation that tank 4 requires. The two adaptive controllers, which identify and invert the actual cross-coupled map online, instead hold tank 4 near setpoint; LCA-RLS-KAN and RBF-NN are statistically indistinguishable there (RBF is marginally ahead). The KAN therefore attains essentially the second-best cross-coupled engagement observed, while winning on every other axis below.

5.2 Actuator Smoothness, Estimation, and Runtime

Table 4 and Fig. 3 report actuator Total Variation (TV) and per-step runtime. Three observations follow. First, LCA-RLS-KAN has the lowest actuator TV of all four controllers; against the near-identical-in-tracking RBF baseline it is 12% and 8% smoother on the two pumps, so the structured symbolic regressor yields a smoother command than a black-box basis under identical adaptation. Second, once the LTV-MPC is given a 3 s prediction horizon (Sec. 4.3) it is notably less smooth than KAN: the warm-started configuration chatters at 1.6–2.1×, and the cold-restarted one at 2.9–3.4×, the KAN's TV, re-planning aggressively at each 3 ms control step. Parameter estimation (Table 5) is comparable across controllers, as the UKF is shared, with the adaptive controllers marginally ahead on γ_1.

Third, on runtime, all controllers execute in 1.2–1.6 ms/step in this efficient reimplementation: the per-step cost is dominated by the shared UKF, and the iterative QP, solved by compiled OSQP, is not the bot-

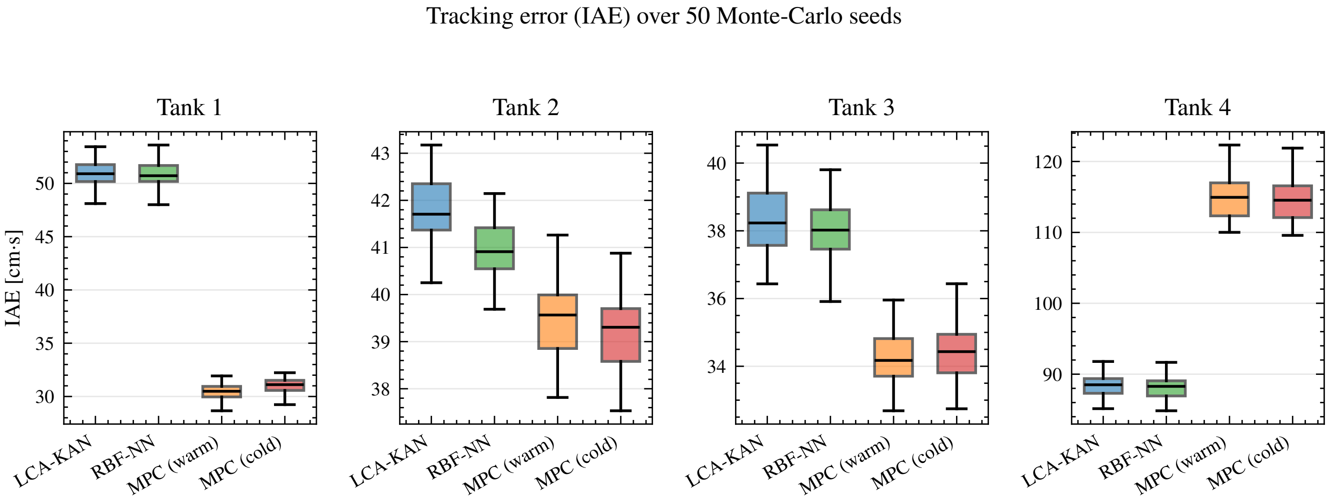

Fig. 2. **Per-tank tracking error (IAE) over 50 Monte-Carlo seeds.** MPC tracks the directly actuated tanks 1–3 tightest but lets the cross-coupled tank 4 deviate further ($\sim 30\%$ higher IAE); the adaptive controllers (LCA-RLS-KAN, RBF-NN) engage the coupled pathway and hold tank 4 near setpoint.

Table 4
Monte-Carlo actuator TV [V] and per-step runtime [ms], median [IQR] over 50 seeds.

Controller	TV, pump 1	TV, pump 2	Runtime [ms]
LCA-RLS-KAN	**1903** [204]	**1328** [284]	**1.20** [0.045]
RBF-NN	2168 [161]	1437 [151]	1.37 [0.066]
LTV-MPC (warm)	4064 [464]	2155 [362]	1.48 [0.027]
LTV-MPC (cold)	6437 [282]	3860 [364]	1.56 [0.176]

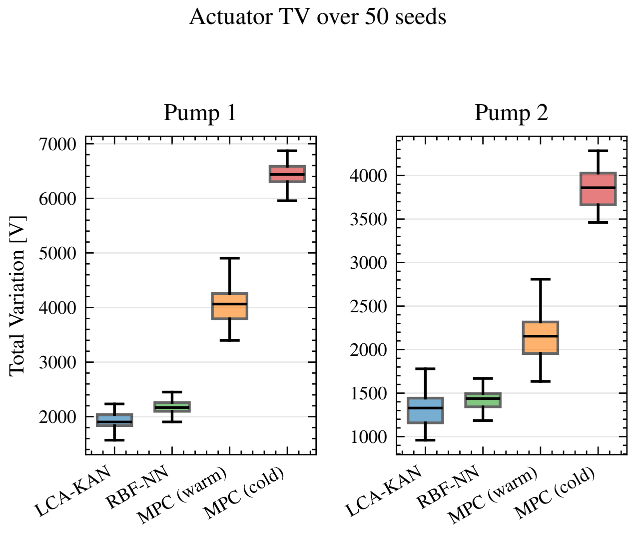

Fig. 3. **Actuator Total Variation over 50 seeds.** LCA-RLS-KAN has the lowest TV of all controllers; given a fair 3 s horizon, both MPC configurations chatter more than the adaptive controllers (warm 1.6–2.1×, cold 2.9–3.4× the KAN's TV).

Table 5
Monte-Carlo parameter-estimation error $|e|$, median [IQR] over 50 seeds.

Controller	$\lvert e_{\gamma_1}\rvert$	$\lvert e_{\gamma_2}\rvert$
LCA-RLS-KAN	0.081 [0.009]	**0.165** [0.007]
RBF-NN	**0.078** [0.008]	0.168 [0.008]
LTV-MPC (warm)	0.096 [0.015]	0.185 [0.018]
LTV-MPC (cold)	0.097 [0.013]	0.185 [0.017]

tleneck a heavier toolchain might make it appear. LCA-RLS-KAN attains the lowest median (1.20 ms), but its decisive advantage is structural rather than average speed: the per-step cost is a fixed, branch-free budget — $\mathcal{O}(p)$ regressor evaluation ($p = 10$), $2p$ multiply–adds of inference, and an $\mathcal{O}(p^2)$ forgetting-RLS update every ten steps — with no solver iterations, tolerance, or factorization, hence a worst-case execution time bounded a priori. The QP cost is instead iteration-count dependent: a warm-started MPC can be made tightly repeatable on a benign run (IQR 0.03 ms here), but the realistic cold-restart case — the one relevant after a reset or fault — is far more variable (IQR 0.18 ms) and admits no a-priori bound. For embedded deployment, where a bounded worst-case execution time matters more than average throughput, this guaranteed determinism is the operative advantage.

5.3 Adversarial Stress Test: Single-Realization Disturbance Recovery

Complementing the Monte-Carlo campaign, we illustrate cross-coupled recovery on a single representative realization in which a step disturbance of 2.0 cm is injected into tank 2 at $t = 30\,\mathrm{s}$, on top of the drift schedule, emulating documented industrial faults (valve obstruction, actuator degradation). No controller is re-tuned. The LTV-MPC trajectories are near-identical for the warm-started and cold-restarted configurations on this scenario (the two differ in actuator TV, not in tracking), and make visible the cross-coupling effect that the aggregate IAE of Sec. 5.1 summarizes.

The analysis focuses on tanks 2–4, the tanks most affected, directly and indirectly, by the disturbance. Both controllers are driven by the same noise realization to isolate differences attributable to the control law. The disturbance response of tank 2 is presented first.

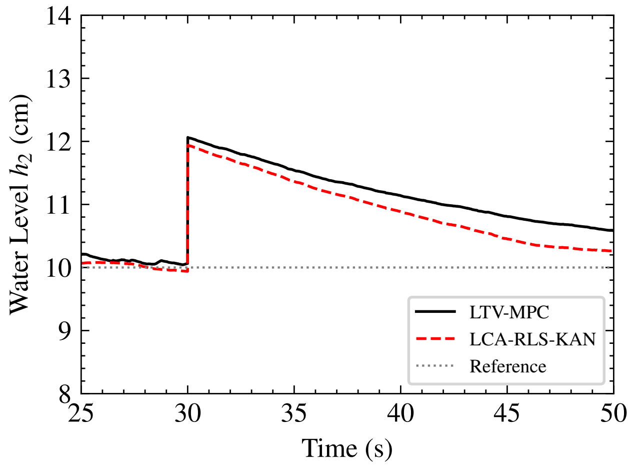

Fig. 4. **Tank 2 level tracking**. At $t = 30\,\mathrm{s}$ a step disturbance of $2.0\,\mathrm{cm}$ is applied, as shown in the plot. The disturbance rejection of the two controllers on the affected tank itself is provided; LCA-RLS-KAN exhibits a steadier recovery compared to LTV-MPC.

As evidenced by Fig. 4, under decaying plant conditions with additional disturbance, LCA-RLS-KAN exhibits faster convergence and reduced steady-state deviation compared to LTV-MPC. The most discriminating adversarial result is tank 3: shared pump actuation forces simultaneous disturbance rejection on tank 2 and h_3 sustainment — a coupled resource allocation problem under model mismatch.

Under a disturbance applied to tank 2, which simultaneously degrades the inflow pathway to tank 3 in the quadruple-tank configuration (Johansson, 2000), LCA-RLS-KAN maintains subsystem coherence, having started the restoration of h_3 at $t \approx 46\,\mathrm{s}$ while LTV-MPC continues to drain h_3 toward zero within the 50-second evaluation window (Fig. 5).

Despite tank 4 being the upstream contributor to tank 2 inflow (Johansson, 2000), LCA-RLS-KAN tracks tank 4 closer to its reference by achieving superior disturbance rejection on the pump 2 subsystem (Fig. 6), indicat-

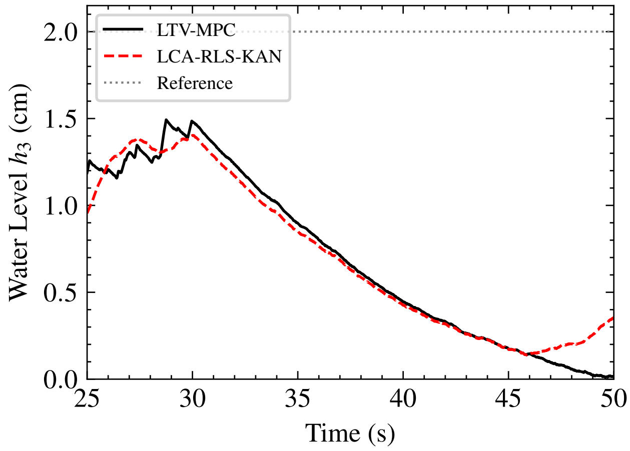

Fig. 5. **Tank 3 level tracking.** As tanks 2 and 3 are interconnected with a single pump, control of tank 3 represents a balancing challenge for both controllers: a necessity to reconcile the setpoint tracking of disturbed tank 2 with the draining of tank 3. The h_3 setpoint (2.0 cm, shared with h_4) is shown for reference; h_3 remains well below it throughout (the large tank-3 steady-state error of Table 6), reflecting the underactuated, non-minimum-phase coupling. Near the end of the window LCA-RLS-KAN begins to restore h_3 while LTV-MPC continues to drain.

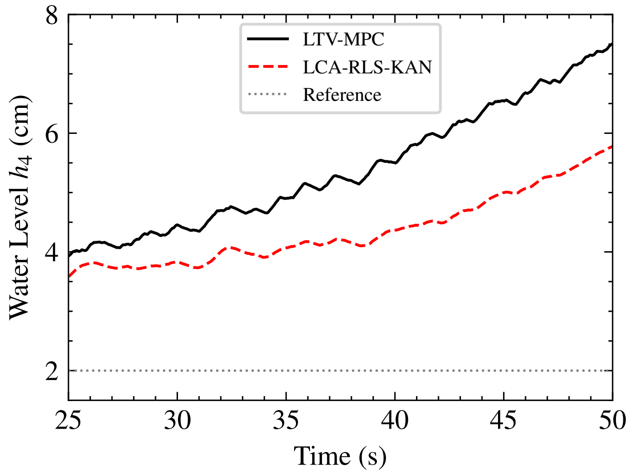

Fig. 6. **Tank 4 level tracking.** Reduced downward flow from tank 3 would increase the voltage needed to maintain the setpoint of tank 1, which in turn causes tank 4 level to rise. Therefore, lower level correlates with higher fidelity of multi-tank control and higher awareness of cross-coupling.

ing stronger coherence of cross-coupled actuation (consistent with Table 4's reduced TV relative to the RBF baseline).

Table 6 provides the IAE and steady-state error (ESS) over the last 5 seconds for both control systems. Time-domain trajectories (Figs. 4, 5, 6) reveal trends not fully captured by ESS values. Thus, simulation campaigns support the claims of efficiency and robustness of the control law.

Table 6
Tracking error metrics for the single-realization disturbance test of Sec. 5 (lower is better).

Controller	**Tank**	**IAE** [cm · s]	**ESS** [cm]
LCA-RLS-KAN	1	56.4	1.92
LTV-MPC		**34.2**	**0.74**
LCA-RLS-KAN	2	**57.3**	**0.33**
LTV-MPC		61.9	0.69
LCA-RLS-KAN	3	58.8	**1.79**
LTV-MPC		**58.0**	1.92
LCA-RLS-KAN	4	**94.3**	**3.33**
LTV-MPC		122.4	4.97

These logs also instantiate the integral band of Lemma 2. The integral coordinate $\hat{d}$ — the low-pass-filtered tracking error accumulated in Algorithm 2 — never exceeds 3.87 cm across the 50 seeds and the disturbance realization, so we set $\bar{d} = 4.0$ cm. With $d^* = 0$ in the idealized setting of Corollary 1 this gives $\bar{d}_{\text{tot}} = 4.0$ cm, and the certified residual radius $\sqrt{(c_2/c_1)(\gamma/\alpha)}\,\bar{d}_{\text{tot}}$ — no smaller than $\bar{d}_{\text{tot}}$ for the conservative Lyapunov constants of the proof — contains the largest steady-state error of the proposed controller (tank 4, 3.33 cm $<$ 4.0 cm). Corollary 1 is therefore consistent with the data rather than vacuous.

The same logs instantiate the lack-of-excitation guarantee of Corollary 2. The empirical feature Gram $G_N = \frac{1}{N}\sum_k \Phi_{\text{KAN}}(\xi_k)\Phi_{\text{KAN}}(\xi_k)^T$ accumulated over the benchmark trajectories has numerical rank seven of ten: regulation to a single setpoint forces the normalized error features onto the level features ($x_5 = c_1 - x_1$, $x_7 = c_3 - x_3$, $x_8 = c_4 - x_4$, with $c_1 = 0.5$ and $c_3 = c_4 = 0.1$ — the $[10, 10, 2, 2]$ cm setpoint scaled by the 20 cm level normalization), collapsing three directions so that the regressor is not persistently exciting. The closed loop is GUUB nonetheless — the valve drift is tracked and the step disturbance rejected (Table 6) — because the certificate of Sec. 4.6 never assumed PE: the covariance bound (Assumption 3) keeps $\Gamma \in [\gamma_{\min} I, \gamma_{\max} I]$ regardless, and Corollary 2 converts that into GUUB of ζ. Only the output-relevant weight combination — the excited, rank-seven subspace — is adapted; the three unexcited directions do not affect the command and need not be identified. Parameter convergence is therefore neither claimed nor required: the single-setpoint regulation of the benchmark is itself a worst case for excitation, and the controller is by design indifferent to it.

5.4 Limitations

Several limitations qualify these results and are stated plainly.

- *Runtime parity, not dominance.* Per-step median runtimes are comparable against an efficient solver (KAN lowest, 1.20 ms); the operative advantage is a branch-free, *a-priori*-bounded worst case, not raw throughput. A warm-started MPC can itself be tightly repeatable on a benign run, so the claim rests on that structural bound rather than on observed variance alone.
- *Frozen basis.* Only the linear weights adapt online; plant changes whose effect lies outside the span of the distilled basis are absorbed merely as a bounded disturbance δ (Lemma 1) rather than learned.
- *Estimator scope.* The estimator ISS underpinning the cascade (Lemma 3) is established regionally on the compact envelope Ω under the verifiable conditions of Assumption 4: the covariance sandwich (i) is read off the logged filter covariance, while uniform observability (ii) and the curvature bound (iii) hold structurally for this plant (levels sensed linearly; valve ratios identifiable under input excitation; $\partial_h^2\sqrt{h}$ bounded since tanks never drain to $h = 0$). The certificate is deterministic, continuous-time, and carries an unscented-linearization bias floor $b_o = \mathcal{O}(\bar{H}\bar{p})$; a global or mean-square statement for the genuine nonlinear filter (Reif et al., 1999; Xiong et al., 2006) and the effect of hardware sampling are left to future work.
- *Residual offset.* The implemented integral state is leaky (low-pass), trading exact integral action for a-priori boundedness and leaving a bounded steady-state offset (Corollary 1). The σ-modified companion of Sec. 4.7 certifies an analogous leakage placed on the bias estimate $\hat{d}$ itself; the implemented filter instead damps the error signal, so its offset is covered by the band of Lemma 2 rather than by Theorem 2.
- *Single benchmark.* Generality is argued through the control-affine class of Sec. 3.1 but demonstrated on one plant; broader empirical validation and hardware-in-the-loop deployment on the target microcontroller are left to future work.

6 Conclusion

This paper presented an LCA-RLS-KAN architecture designed to bridge embedded feasibility and advanced nonlinear control. A Lyapunov-based analysis established ISS and GUUB of the tracking and weight errors, with practical tracking accuracy in the ideal (disturbance-free) case shown to be governed directly by the band bounding the integral state, and robustness under lack of PE proven. The theoretical guarantees explicitly account for bounded estimation error and a bounded, leaky integral action.

The proposed controller was compared over a 50-seed Monte-Carlo on the quadruple-tank benchmark under an identical adversarial drift schedule, against a black-box RBF adaptive baseline and LTV-MPC in warm- and cold-restarted configurations.

Across the campaign, LCA-RLS-KAN matched the aggregate tracking of LTV-MPC and the RBF baseline while engaging the cross-coupled, non-minimum-phase actuation that MPC under-served — holding the coupled tank approximately 23% closer to setpoint, despite the LTV-MPC baseline possessing a 3 s prediction horizon. It also attained the lowest actuator total variation of all four controllers — a prediction-horizon MPC chatters more, not less, than the distilled symbolic regressor — and the lowest median per-step cost at a branch-free, worst-case-bounded evaluation. Under a single-realization step disturbance, LCA-RLS-KAN preserved subsystem coherence and restored cross-coupled trajectories more effectively within the observed horizon.

Overall, the proposed framework combines lightweight adaptive control structure with interpretable nonlinear symbolic distillation via Kolmogorov-Arnold Networks (KANs), yielding a verifiable and computationally efficient regressor-based control law suitable for constrained embedded systems.

References

Alvarado, I., Limón, D., Muñoz de la Peña, D., Maestre, J.M., Ridao, M.A., Scheu, H., Marquardt, W., Negenborn, R.R., De Schutter, B., Valencia, F., Espinosa, J., 2011. A comparative analysis of distributed MPC techniques applied to the HD-MPC four-tank benchmark. Journal of Process Control 21, 800–815.

Arnold, V.I., 1957. On functions of three variables, in: Doklady Akademii Nauk, Russian Academy of Sciences. pp. 679–681.

Astrom, K.J., 1987. Adaptive feedback control. Proceedings of the IEEE 75, 185–217.

Besançon, G., 2000. Remarks on nonlinear adaptive observer design. Systems & Control Letters 41, 271–280.

Borrelli, F., Bemporad, A., Morari, M., 2017. Predictive control for linear and hybrid systems. Cambridge University Press.

Boyd, S., Vandenberghe, L., 2004. Convex optimization. Cambridge University Press.

Brunton, S.L., Proctor, J.L., Kutz, J.N., 2016. Sparse identification of nonlinear dynamics with control (SINDYc). IFAC-PapersOnLine 49, 710–715.

Cruz, G., Renczes, B., Runacres, M.C., Decuyper, J., 2025. State-space Kolmogorov Arnold networks for interpretable nonlinear system identification. IEEE Control Systems Letters .

Diamond, S., Boyd, S., 2016. CVXPY: A Python-embedded modeling language for convex optimization. Journal of Machine Learning Research 17, 1–5.

Fortescue, T., Kershenbaum, L.S., Ydstie, B.E., 1981. Implementation of self-tuning regulators with variable forgetting factors. Automatica 17, 831–835.

Gatzke, E.P., Meadows, E.S., Wang, C., Doyle, III, F.J., 2000. Model based control of a four-tank system. Computers & Chemical Engineering 24, 1503–1509.

Ge, S.S., Hang, C.C., Lee, T.H., Zhang, T., 2013. Stable adaptive neural network control. volume 13. Springer Science & Business Media.

Hose, H., Gräfe, A., Trimpe, S., 2024. Parameter-adaptive approximate MPC: Tuning neural-network controllers without retraining, in: 6th Annual Learning for Dynamics & Control Conference, PMLR. pp. 349–360.

Ioannou, P.A., Sun, J., 1996. Robust adaptive control. volume 1. PTR Prentice-Hall Upper Saddle River, NJ.

Jazwinski, A.H., 1970. Stochastic Processes and Filtering Theory. Academic Press.

Jerez, J.L., Goulart, P.J., Richter, S., Constantinides, G.A., Kerrigan, E.C., Morari, M., 2014. Embedded online optimization for model predictive control at megahertz rates. IEEE Transactions on Automatic Control 59, 3238–3251.

Johansson, K.H., 2000. The quadruple-tank process: A multivariable laboratory process with an adjustable zero. IEEE Transactions on Control Systems Technology 8, 456–465.

Julier, S.J., Uhlmann, J.K., 2004. Unscented filtering and nonlinear estimation. Proceedings of the IEEE 92, 401–422.

Kalman, R.E., 1960. A new approach to linear filtering and prediction problems. Transactions of the ASME–Journal of Basic Engineering 82, 35–45.

Khalil, H.K., Grizzle, J.W., 2002. Nonlinear systems. 3 ed., Prentice-Hall Upper Saddle River, NJ.

Kim, T., Girard, A., Kolmanovsky, I., 2025. CIKAN: Constraint Informed Kolmogorov-Arnold networks for autonomous spacecraft rendezvous using Time Shift Governor, in: Proceedings of the 7th Annual Learning for Dynamics & Control Conference, PMLR. pp. 1115–1126.

Kolmogorov, A.N., 1957. On the representations of continuous functions of many variables by superposition of continuous functions of one variable and addition, in: Dokl. Akad. Nauk USSR, pp. 953–956.

Krstic, M., Kokotovic, P.V., Kanellakopoulos, I., 1995. Nonlinear and adaptive control design. John Wiley & Sons, Inc.

Labbe, R.R., 2014. Kalman and Bayesian Filters in Python. Chapman & Hall/CRC. URL: `https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python`.

Lewis, F., Jagannathan, S., Yesildirak, A., 2020. Neural network control of robot manipulators and non-linear systems. CRC Press.

Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., Hou, T.Y., Tegmark, M., 2024. KAN: Kolmogorov-Arnold networks. arXiv preprint arXiv:2404.19756 .

Marino, R., Tomei, P., 1992. Global adaptive observers for nonlinear systems via filtered transformations. IEEE Transactions on Automatic Control 37, 1239–1245.

Mayne, D.Q., Rawlings, J.B., Rao, C.V., Scokaert, P.O., 2000. Constrained model predictive control: Stability and optimality. Automatica 36, 789–814.

Narendra, K.S., Annaswamy, A.M., 2012. Stable adaptive systems. Courier Corporation.

Plackett, R.L., 1950. Some theorems in least squares. Biometrika 37, 149–157.

Reif, K., Günther, S., Yaz, E., Unbehauen, R., 1999. Stochastic stability of the discrete-time extended kalman filter. IEEE Transactions on Automatic Control 44, 714–728.

Salkimbayev, A., 2026. Symbolic Kolmogorov–Arnold networks as a constant-time alternative to linear time-varying model predictive control for embedded control of non-minimum phase systems. Engineering Applications of Artificial Intelligence. Under review.

Sanner, R.M., Slotine, J.J.E., 1991. Gaussian networks for direct adaptive control, in: 1991 American Control Conference, IEEE. pp. 2153–2159.

Skogestad, S., Postlethwaite, I., 2005. Multivariable feedback control: analysis and design. John Wiley & Sons.

Slotine, J.J.E., Li, W., 1991. Applied nonlinear control. Prentice-Hall Englewood Cliffs, NJ.

Sontag, E.D., 1989. Smooth stabilization implies coprime factorization. IEEE Transactions on Automatic Control 34, 435–443.

Stellato, B., Banjac, G., Goulart, P., Bemporad, A., Boyd, S., 2020. OSQP: An operator splitting solver for quadratic programs. Mathematical Programming Computation 12, 637–672.

Wan, E.A., Van Der Merwe, R., 2000. The Unscented Kalman Filter for nonlinear estimation, in: Proceedings of the IEEE 2000 Adaptive Systems for Signal Processing, Communications, and Control Symposium (Cat. No.00EX373), IEEE. pp. 153–158.

Xiong, K., Zhang, H.Y., Chan, C.W., 2006. Performance evaluation of UKF-based nonlinear filtering. Automatica 42, 261–270.

Zhang, Q., 2002. Adaptive observer for multiple-input-multiple-output (MIMO) linear time-varying systems. IEEE Transactions on Automatic Control 47, 525–529.